

**Succession of soil borne diseases of chickpea and their
management through host plant resistance**

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In

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(PLANT PATHOLOGY)**

By

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CERTIFICATE – I

This is to certify that the thesis entitled, **Succession of soil borne diseases of chickpea and their management through host plant resistance** submitted in partial fulfilment of the requirement for the degree of **“MASTER OF SCIENCE IN AGRICULTURE”** in plant pathology of the Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur is a record of the bonafide research work carried out by **Mr. Akhilesh Jagre** under my guidance and supervision. The subject of the thesis has been approved by the Student’s Advisory Committee and the Director of Instruction.

No part of the thesis has been submitted for any other degree or diploma (Certificate awarded etc.) or has been published/published part has been fully acknowledged. All the assistance and help received during the course of investigation has been duly acknowledged by him.

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CERTIFICATE – II

This is to certify that the thesis entitled **Succession of soil borne diseases of chickpea and their management through host plant resistance** submitted by **Mr. Akhilesh Jagre** to the Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur, in partial fulfilment of the requirements for the degree of **“Master of Science in Agriculture”** in the Department of **Plant Pathology** has been, after evaluation, approved by the External Examiner and by the Student’s Advisory Committee after an oral Examination on the same.

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INTRODUCTION

Chickpea (*Cicer arietinum* L.) is an important pulse crop, which belongs to Leguminosae family, ranking third after dry beans (*Phaseolus vulgaris* L.) and dry peas (*Pisum sativum* L.) (Dhar and Gurha, 1998). The Kabuli and Desi chickpea is grown throughout the world with different names i.e. Chickpea (UK), Garbanzo (Latin America), Bengal gram (India), Hommes, Hamaz (Arab world), Shimbira (Ethiopia) and Nohud and Loblebi (Turkey). Chickpea is self-pollinated rabi crop, upto 1% cross pollinated (Smithson *et al.*, 1985; Singh, 1987).

In India Chickpea is grown in 8.57 m ha with 7.8 m. tones production and 900-920 kg/ha productivity. It is grown in over 45 countries in all continents of the world. It provides a high quality protein to the people in developing countries. Chickpea is valued for their nutritive seeds with high protein content, 25.3 - 28.9 per cent after dehulling (Hulse, 1991) carbohydrate 61.5 per cent, fat 4.5 per cent and vitamin 2.44 per cent.

The world's total production of chickpea is around 8.5 m tones annually and is grown over 11.2 m. ha. of land approximately. The desi chickpea contribute to around 80 per cent of the total production. India is the largest producer of chickpea contributing around 70 per cent (6.2 m. tones) of the world's total production. Apart from India, Turkey (7%), Pakistan (5%), Iran (3%), Mexico (3%), Australia (2%), Canada (2%) and Ethiopia (2%) were found to be the other major chickpea producing countries of the world (Agri. Statistics at a glance, 2010-11).

Chickpea comprises of two main types; one desi type with small and brown seed accounts for nearly 90 per cent and kabuli type with bold and cream-colored seed, grown in around 10 percent area. In India the major chickpea growing states are Madhya Pradesh, Rajasthan, Maharashtra, Gujarat, Andhra Pradesh, and Karnataka. Madhya Pradesh ranks 1st in both area (2.9 m. ha) and production 2.6 (m. tones) and the average productivity is estimated 931 kg / ha. (Agri. Statistics at a glance, 2009-10). Madhya

Pradesh has the privilege of growing a wide range of chickpea varieties, which categories in three type viz; desi, gulabi and kabuli and grown as sole or mixed crop with wheat, mustard, lentil etc. Chickpea is affected by several seed, soil and air borne diseases which is responsible for lowering its yield. Soil borne pathogens like *Sclerotium rolfsii* (Collar rot), *Fusarium oxysporum* f. sp. *ciceri* (Vascular wilt) and *Rhizoctonia bataticola* (Dry root rot) are responsible for causing diseases from seedling to flowering and pod formation stage.

Collar rot caused by *Sclerotium rolfsii* Sacc. is an important soil borne disease of chickpea, causing seed rot and seedling mortality in the initial stage of crop growth up to 45 days. The mortality ranges from 54.7- 95 per cent in India (Kotasthane *et al.*, 1976) causing significant reduction in plant population.

The pathogen is an omnivorous soil borne fungus, parasitizing root and collar region of the plant thus, producing wilt like symptoms. Generally the disease is severe in loamy soil regions and more prevalent in soybean- paddy based cropping system.

It is generally observed when soil moisture is high and temperature is warm at the seedling stage. Affected seedlings turn yellow and die. The seedlings generally collapse and show rotting at the collar region and below. Available literature reveals that different screening techniques have been employed for screening large number of germplasm for collar rot resistance, but sources are scanty.

Chickpea grows best as a post-monsoon cool-season crop in semi-arid regions of the subcontinent. It takes 80 to 170 days to mature. Optimum conditions for growth include 21 to 29°C nights and 18 to 26°C days' temperature with 600-1000 mm annual rainfall (Muehlbauer *et al.*, 1988; Duke, 1981).

Fusarium wilt is a serious disease, especially in low rainfall areas, where weather conditions are favourable for disease development. It has been reported from 33 countries of the world (Nene *et al.*, 1996)

The first report of occurrence of root rot in chickpea along with wilt was made by Padwick (1948). The species *Rhizoctonia bataticola* responsible for serious diseases (Mukharji and Bhasin, 1986). Gupta *et al.*, (1983) reported incidence of root rot ranging 3.58 to 20.63 percent in 30 villages of Northern Madhya Pradesh. The disease is more severe at pod formation and grain filling stage especially under stress condition and causes forced maturity resulting reduced seed size.

Hence, the present study was undertaken with the following objectives, which will help in determining the virulence of various isolates of *Rhizoctonia bataticola* on chickpea and in identifying the host resistance. Looking to the importance of soil borne pathogens which causes losses at different stages of crop growth hence studies were conducted to know the succession of the chickpea diseases and their relation with the temperature and moisture condition besides crop growth stage. Management through host plant resistance will offers an opportunity in resistance breeding programs. The present investigations were conducted with the following objectives;

Objectives:

- To know the succession of soil borne diseases of chickpea with regard to crop growth stage and its relation with the weather data.
- To identify the host plant resistant sources against collar rot, vascular wilt and dry root rot.

REVIEW OF LITERATURE

Chickpea is most extensively grown pulse crop in India. It is attacked by air borne, seed borne, and soil borne diseases.(Chattopadhyay *et al.*,2001) more than 75 pathogen have been reported to infect chickpea (Nene *et al.*, 1984)

Among soil borne diseases collar rot (*Sclerotium rolfsii*), Vascular wilt (*Fusarium oxysporum* f.sp. *ciceri*) and dry root rot (*Rhizoctonia bataticola*) are most important diseases causing heavy losses from seedling to podding stage and disease severity depends upon the temperature and moisture condition.

2.1 Historical

2.1.1 Collar rot

Peter Henary Rolfs (1892), first published a description of a new disease on tomato, from floride and recorded losses upto 70 percent. Later Saccardo named the fungus as *Sclerotium rolfsii* in 1911, which continues to be a problem in a variety of crops when conditions are warm, humid and rainy. Today, this disease occurs all over the world in the equatorial zone between 45⁰ N and 28.5⁰ S latitude. Later in 2000 Ansari *et al.* showed that *S. rolfsii* is one of the serious polyphagous pathogen causing disease in as many as 500 plant species.

2.1.2 Vascular wilt

Chickpea wilt was first reported by Butler in his book entitled “Fungi and disease in plants” in 1918. In 1923, Mackerel working in Burma considered the disease to be soil borne and wilted samples yielded species of *Fusarium* and *Rhizoctonia*.

McRae (1929), Narsimhan (1929) and Padwick (1939) experimentally proved that wilt disease is caused by species of *Fusarium*. Padwick (1940) identified the casual agent of Fusarium wilt as *Fusarium orthoceras* and Wollr var. *ciceri*. He also reported that symptoms due to *Opercullela padwickii* and virus yellows can be confused with wilting. In 1935,

Dastur (1935) found *Rhizoctonia bataticola* to cause wilt symptoms in plants and he called the disease *Rhizoctonia* wilt, his description of symptoms and field pattern of disease incidence was almost identical to those of typical wilt caused by *Fusarium oxysporum* f. sp. *ciceri* but he failed to prove the pathogenicity and lastly concluded that the wilt was due to physiological reasons. He called it as physiological wilt.

In 1984, Reddy and Singh while conducting studies on wilt reported that chickpea germplasm resistant to Ascochyta blight and *Fusarium* wilt are limited in existing chickpea species so it is necessary to search out new sources of resistance to this disease.

In addition to *Fusarium* sp. (*F. oxysporum* and *F. solani*) other fungi such as *Rhizoctonia* sp. and *O. padwickii* have found associated with Bengal gram wilt (Anon 1972). The disease is a vascular pathogen that travel in seed and soil consequently and it is difficult to handle it by use of chemicals and through crop rotation (Kraft *et al.*, 1994 and Pande *et al.*, 2007)

2.1.3 Dry root rot

The first mention of a plant disease, which may be referred with, certainly to *Rhizoctonia* as a causal agent is an important paper by Du Hamel (1728) read before the Paris Academy. In this paper he gave a careful description of a fungus disease of saffron (*Crocus sativus*) occurring in France. He named the fungus *Tuberorides*.

De Candolle (1815) discovered a similar disease on alfalfa. In his taxonomic description he was first person who gave the name for the fungus. Later, after careful study of the characteristics and parasitism of the species, he established (1815) the genus *Rhizoctonia*.

In 1924, Small described a *Rhizoctonia*, which develop black sclerotia of variable size and shape in the diseased roots, many woody plants from Uganda and he named it *Rhizoctonia lamrllifera*. Butler (1925), however considered that the fungus should be regarded as *Rhizoctonia* and made the combination *Rhizoctonia bataticola* (Taub.) Butler to a *Sclerotium* causing a root rot of cotton, french beans and cowpeas.

2.2 Occurrence and losses

2.2.1 Collar rot

Rolf (1982) reported a serious trouble of tomato in Florida first time. Later the fungus species was identified as *Sclerotium rolfsii* by Saccardo (1911). Information on the losses caused by this fungus is inadequate. It caused mortality 20-60% foliage in India (Sikarwar, 2003).

2.2.2 Vascular wilt

The disease has been reported from several countries including India, Bangladesh, Burma, Ethiopia, Mexico, Pakistan, Syria and Tunisia (Nene *et al.*, 1984). The disease is widely prevalent throughout India being very destructive in parts of Uttar Pradesh, Madhya Pradesh, Bihar, Punjab, Maharashtra and Eastern India.

Grewal *et al.*, (1974) reported 2 to 5 per cent yield loss every year in India, but it could go as high as 60 per cent. Similarly, early wilting reduced the seed number per plant and caused more yield losses than late wilting (Haware and Nene, 1980). The yield loss is reported up to 10 per cent in India (Singh and Dahiya 1973). However, in favourable conditions the crop destroys completely.

An annual loss of 12 million rupees approx US dollar 1 million has been reported from Pakistan by Sattar *et al.*, (1953). Nema and Khare (1973) observed damage up to 61 per cent at seedling stage and 43 per cent at flowering stage from this disease in Madhya Pradesh.

2.2.3 Dry root rot

The first report of the occurrence of root rot in chickpea along with wilt was made by Padwick (1948). The incidence of root rot ranging from 3.58 to 20.63 per cent in 30 villages of northern Madhya Pradesh was reported by Gupta *et al.*, (1983).

Tripathi and Sharma (1983) studied the incidence of chickpea dry root rot (*Rhizoctonia bataticola*) in Southern Haryana and reported that the incidence was high from late October to mid November which was described in December and January and then again increased in

the months of February and March. They also noted that seed sown at a depth of 15 cm developed a less root rot than that sown at 7.5 cm.

The *Rhizoctonia bataticola*, responsible for serious diseases like stem blight, foliage blight, root rot, seedling blight, charcoal rot, wilt, collar rot, dry rot, pod and seed rot (Raut and Ingul, 1989) in several crops such as soybean, gram mung, urid, fruit trees and many vegetables (Mukharji and Bhasin, 1986). This *sclerotial* fungus is of wide geographical distribution ranging from Formosa, Philippines, India, Ceylon, East Africa, Palestine and Egypt to the West Indies and the Eastern United State (Ashby, 1927). Significant reduction (25-70%) in chickpea production is caused by this disease in India (Ahmed *et al.*, 1986 and Pandey and Singh, 1990).

2.3 Symptoms

2.3.1 Collar rot

Collar rot caused by *sclerotium rolfsii* Sacc. is an important soil borne disease of chickpea, causing seedling mortality in the initial stage of crop growth. The disease was found to occur mainly at seedling stage of crop growth (15 – 45 days). The affected plants turn yellow and die. The seedlings found to collapsed and showed rotting at the collar region and below. The affected plants pulled very easily. White mycelial growth adhered to collar region along with mustard like sclerotia was observed.

2.3.2 Vascular wilt

Several workers have described symptoms of the disease as they occur in different countries. Narsimhan (1929), Erwin (1958), Chauhan (1962), Westurland *et al.*(1974) reported that affected seedlings showed no external rotting of the root system. Mossahebi (1968) described the wilt symptoms as general yellowing of the leaves and discoloration of vascular elements. Echandi (1970) stated that main symptoms were wilting and yellowing of foliage and necrosis of roots. Nene *et al.* (1979) described the symptoms of chickpea wilt as sudden drooping of leaves and petioles with external rotting of roots.

Nema and Khare (1973) reported that chickpea cultivation in madhya Pradesh was greatly threatened by soil borne diseases generally appearing at two phases of growth, early seedling stage and flowering stage, wilting was due to *Fusarium oxysporum* f.sp.*ciceri*, dry root rot caused by *Rhizoctonia bataticola* and *Sclerotium rolfsi* resulted in collar rot.

2.3.3 Dry root rot

Rhizoctonia bataticola is a soil borne fungus/pathogen. Several workers have reported various types of symptoms produced on plants with varying intensity of the disease in different crops and different regions. Narsimhan (1929) observed that plants infected with *R. bataticola* initially showed drooping of leaves followed by wilting. The main roots of such plants show blackening of finer roots.

According to Mehrotra and Aneja (1990) the first symptom of the disease is yellowing of the leaves within a day or two such leaves drop and in the course of the next two or three days whole plant dies off. The leaves and stem of the affected plants are usually straw coloured. The taproot is dark and quite brittle in dry soil.

Demiro *et al.*, (1994) observed that dry root rot disease of gram is also becoming a serious disease affecting productivity of this crop. They isolated *Rhizoctonia bataticola* from plants and seeds of chickpea cultivars Azizive 94 and tested pathogenicity.

2.4 Pathogenicity

Grewal *et al.*, (1974) made several isolation from wilted chickpea plants in pathogenicity test *Fusarium oxysporum* was found to virulent and caused 82 to 100 percent mortality of inoculated within 20 days.

Zote *et al.*, (1990) tested the pathogenicity of eight isolates of *Fusarium oxysporum* f. sp. *ciceri* and found that all of them induce wilt symptoms.

2.5 Host resistance

The best method of control is through the use of resistant cultivars. Efforts have been made at various institutes in the country to locate the sources of resistance. Many reports on the identification of resistance to

vascular wilt have appeared in the literature during the last 85 years. These reports were based on observations made during natural epidemics while several were based on artificial inoculation tests in the field or in green house (Singh, 1974)

Mathur *et al.*, (1975) suggested method to detect the seed borne fungi with slight modification.

Agrawal *et al.*, (1976) reported susceptibility of lentil plants to *Sclerotium rolfsii* at different stages of growth in pot tests at Jabalpur. Seedling mortality was highest when inoculated just after sowings, after that decrease with plant age.

Kotasthane *et al.*, (1976) reported seedling mortality in chickpea ranging from 54.7 to 95% in the initial stage of crop growth up to 30-35 days.

Mishra *et al.*, (1977) screened 1542 cultivars and strains of chickpea against wilt at Kanpur. Out of which six strains namely 1189-1, P-1424-1, P-6217, GW-1, GW-3 and GW-9 were found resistant (5 per cent mortality)

According to Haware *et al.*, (1978) the fungus *Fusarium oxysporum* f.sp.*ciceri* is primarily soil borne but few reports indicated that it can be transmitted through seeds. The fungus survives in soil in the absence of host for more than six years. He also reported that intensity of wilt fungus was quantified in the seeds of chickpea cultivar K 850 collected from wilted plants as per standard procedures of seed health testing.

Gurha and Dubey (1983) found possible sources of resistance in chickpea against *S. rolfsii* (H 208, K 902, K 850, ICC 5105, ICC 6161) as tolerant.

Karat *et al.* (1985) observed collar rot disease is important in rainfed crops but is less severe under irrigation where it is easily controlled by alterations to the schedule. Of 8 cultivars assessed in the field only NP-3 remained free from infection.

Nene (1988) has developed three screening techniques viz. sick plot, pot culture and water culture. The pot and water culture techniques are usually adopted to supplement field screening test.

Nene *et al.*, (1989) identified broad based stable resistant lines to wilt and root rots, on the basis of multi location evaluation lines viz ICC11550, ICC11551, ICC9032, ICC 9025, ICC2862.)

Chittle *et al.* (1990) reported RSG 130 and RSG 132 as resistant and RSG 191 as moderately resistant against collar rot.

Reddy *et al.* (1990) reported chickpea varieties resistant to wilt and collar rot. The evaluation was done at ICRISAT centre during season 1985-86 to 1988-99 with a row of susceptible cultivar JG 62 between every second row of test lines. Mortality of JG 62 in all four season was 100 percent. The lines avrodhi and ICC 48 which were tested in all the season had a mortality of less than 20 per cent whereas, line GCP 6 which was tested in 3 season also showed mortality less than 20 per cent.

Sugha *et al.*, (1991) developed techniques for screening chickpea varieties against collar rot caused by *S. rolfsii* where one wheat grain fully covered with mycelium of the pathogen at the collar region of a 7 days old seedling has been placed in a pot containing sterilized garden soil. Seedling mortality begin in with 48 hrs.

Salem *et al.*, (1991) screened chickpea accessions for their reaction to the root rot /wilt disease complex in Egypt. Giza 88, L-606 and L-1164 were found resistant to disease.

Bhatt (1993) observed that BGM 416, BG 416, ICC 1376 and IC 11314 were resistant to root rot. Resistant cultivars had greater number of total roots during early growth phases.

Gupta (1995) screened 300 diverse chickpea lines in a multiple disease sick plot and identified seven lines as resistant having <10 per cent mortality.

Halila and Strange (1997) screened a total of 1915 Kabuli chickpea lines in a wilt sick plot and complete resistance was observed in only 110 lines.

Yadav *et al.*, (2000) evaluated eighty one chickpea genotypes for their reaction to wilt. Out of these 10 genotypes were resistant to (JG-22, JG 33, KPG 316 and SAKI 93130) showing a stable resistant reaction and being suitable for use in breeding programmes.

Gupta (2001) made studies on the reaction of chickpea genotype against collar rot by using potting medium mixed with wheat straw, sand medium @ 50g inoculum per 2 kg sterilized sand + soil (1:1) mixture, further among 423 desi and kabuli chickpea genotypes against collar rot in multiple disease sick plot revealed 5 genotypes of desi and 4 of kabuli types as resistant and 39 as moderately resistant .

Gangwar *et al.*, (2002) screened 35 chickpea cultivars for resistance to dry root rot caused by *Rhizoctonia bataticola* in a field experiment. All the cultivars exhibited disease symptoms, only 10 cultivars (ICC 2644, 10384, 10630, 112244, 11332, ICCL 81002, 810810, ICC 12263, 12441 and ICCV 90254) were resistant whereas 5 genotypes (GCP 9504, Phule G 96020, 96105, 96313 and GL 91059) were moderately resistant to dry root rot.

Gurha *et al.*, (2002) screened 570 chickpea genotypes for resistance to isolate (Race 2) of *F. oxysporum* and reported 21 cultivars as stable resistance against *Fusarium* wilt..

Prajapati *et al.*, (2003) while working on resistant sources of chickpea against dry root rot (*Rhizoctonia bataticola*) evaluated forty-five chickpea cultivars for resistance to dry root rot. Among these, H 355, JG 315, Pusa 212, G 543, ICC 32, BG 244, Phule G 5, ICC numbers 2644, 10384, 11224, 10630 and 11332, ICCV 10, ICC numbers 12441 and 12263, 81010 and 81002, Avrodhi, ILC 702 and ILC 200 cultivars were rated as resistant. Five cultivars were moderately resistant and 9 were tolerant.

Pande *et al.*, (2004) evaluated twenty-nine chickpea germplasm accessions, 10 cultivars and 8 advanced breeding lines for resistance to dry root rot, caused by *Rhizoctonia bataticola* under *in vitro* conditions. They observed that one germplasm accession (ICC

14395), a cultivar (ICCV 2) and advanced breeding line were resistant to dry root rot. Of the remaining lines 22 were moderately resistant, 19 susceptible and 2 highly susceptible line (BG 212 and ICC 12267) used as the control.

Hussain *et al.*, (2005) evaluated fifty-seven chickpea germplasm accessions under greenhouse conditions to identify sources of genetic resistance against collar rot disease caused by *Sclerotium rolfsii*. Only one genotype, FLIP 97-174C, was found highly resistant, while five genotypes, FLIP 00-55C, ICC-4936, ICC-13051, ICC-12961 and ICC-14911, were identified as resistant. Twenty genotypes were found moderately resistant (tolerant) while the remaining 31 were susceptible to highly susceptible.

Gupta and Babbar (2006) screened 271 genotypes comprising of Desi and Kabuli chickpea against wilt pathogen in multiple disease sick plot at Jabalpur. Genotype H099-264, PG – 9425-5, PG – 9425-9, GF 240, MPJ6-98-9032 and MPJG – 98-11551 (Desi) and HK 0029, PG – 97-313, HK 94-134 (Kabuli) showed resistance to wilt.

Gupta and Babbar (2006) screened 233 desi and kabuli chickpea genotypes with along 38 promising lines against collar rot (*Sclerotium rolfsii*), dry root rot (*Rhizoctonia bataticola*) and vascular wilt (*Fusarium oxysporum* f. sp. *ciceris*) in multiple disease sick plot at Jabalpur, Madhya Pradesh, India. Each genotype was alternated with a single row of susceptible control (JG 62) and resistant control (JG 63). Genotypes H 99-264, PG 9425-5 and PG 9425-9 (desi) and HK 00297 and PG 97-313 (kabuli) exhibited resistance against all the 3 pathogens. WCG 97-29, WCG 97-37, IPC 97-72 and H 00-256 (desi) showed less than 10% mortality for collar and dry root rots. These genotypes are valuable sources of resistance and may be exploited for utilization in breeding programmes for developing high-yielding cultivars with multiple disease resistance.

Hussain *et al.*, (2006) studied the effects of inoculum potential, age of the seedling (6, 12, 18, 24 and 30 days old), and soil type (clay, clay loam, sandy loam and sandy) on the incidence of collar rot disease in chickpea caused by *Sclerotium rolfsii*. Among the inoculation techniques, the use of

inoculum in the form of wheat grains fully impregnated with mycelium of the pathogen was found to be simple, feasible and highly effective. Younger seedlings were more susceptible, and the susceptibility decreased with the increase in the age of seedlings.

Chaudhary *et al.*, (2007) screened 196 chickpea germplasm lines/cultivars for resistance to wilt disease in a wilt sick plot. None of the test line was found immune or highly resistant.

Pande *et al.*, (2007) evaluated recently developed 117 desi chickpea genotypes at ICRI SAT India against *Fusarium* wilt. Sick plot technique has been widely used and found most efficient.

Abida Akram *et al.*, (2008) evaluated ninety-eight chickpea germplasm accessions received from International Centre for Agricultural Research in the Dry Areas (ICARDA), Syria and NARC, Pakistan there germplasm were evaluated under greenhouse conditions where temperature, ranged from 8-24 degrees C and humidity was maintained above 80% by sprinkling fresh water during the year 2006 to identify sources of genetic resistance against collar rot disease incited by the fungus *Sclerotium rolfsii*. Out of 98 germplasm accessions only 5 genotypes viz., FLIP 97-132C, FLIP 97-85C, FLIP 98-53C, ILC-5263 and NCS 9905 exhibited highly resistant response to disease while 9 genotypes viz., FLIP 96-153C, FLIP 97-129C, FLIP 97-172C, FLIP 97-185C, FLIP 98-227C, FLIP 98-107C, FLIP 98-230C, ILC-182 and NCS 9903 displayed resistant reaction. Twenty five genotypes displayed moderately resistant to tolerant response while the remaining were susceptible to highly susceptible to this disease.

Dua *et al.*, (2008) reported resistance sources against individual as well as multiple diseases in chickpea were carried out during 2004-05 from the germplasm available at the NBPGR New Delhi India. The numbers of promising accessions having resistance against individual disease were 19 for wilt, 11 for dry root rot, 24 for collar rot. ICS 327060, 327062, 327063, 327073, 327074, 327100, 327112, 327190, 327220, 327259, 327336, 327679, 327777, 327779 and 327975 were identified as promising as these

had combined resistance against wilt, dry root rot, collar rot, wet root rot and foot rot.

Jayalakshmi *et al.*, (2008) conducted a pot culture experiment to screen chickpea promising entries (12+1 susceptible check JG-62) during rabi season of 2004-2005 and 2005-2006 against dry root rot. Out of 12 genotypes tested against root rot disease 4 were found resistant, 2 were moderately susceptible, 3 were susceptible and 3 were highly susceptible to dry root rot disease.

Gupta and Mishra (2009) evaluated 120 lines in disease sick fields for 3 consecutive years and 32 entries performed consistent resistant reaction to collar rot. Twelve accessions were found free from collar rot during the testing years under high disease pressure.

Gupta *et al.*, (2012) screened for resistance against *Rhizoctonia bataticola* causing dry root rot in chickpea. The susceptible cultivar BG 212 showed 100 per cent mortality *i.e.* rating 9 in 1-9 scale. The studies led to conclusion that out of 170 accessions, 68 genotypes exhibited resistant reaction (<10% mortality), out of which 26 are the promising lines namely (JG1-14, 2-125, 2-4-110, 14-11, 14-10, 2001-13, 2001-13, 2001-18, 2001-80, 2001-115, 2002-20, 2003-95, 2003-14-16, 2004-110, 210, 9605, 1-9, 99-115, 2001-04, 2003-14-2, JG 2000-07, JSC 37, MPJG89-11551, MPJG 89-9023, CSJ 592 and Rajas) from JNKVV, Jabalpur. These lines further evaluated for their performance in sick field for three consecutive years and revealed six lines *viz.*, JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas as resistant exhibiting <10 per cent mortality, however 14 lines showed moderately resistance reaction.

MATERIAL AND METHODS

Present investigations were undertaken to study the **Succession of soil borne diseases of chickpea and their management through host plant resistance**. The materials used and methods followed are described below:

A. Succession of soil borne diseases of chickpea

To know the succession of soil borne diseases of chickpea occurring from seedling to flowering and pod formation stages, weekly observations on plant mortality due to collar rot (*Sclerotium rolfsii*), vascular wilt (*Fusarium oxysporum*) and dry root rot (*Rhizoctonia bataticola*) were recorded and correlated with the weather data (Nov. 2011 – March 2012).

B. MATERIAL

3.1 General

Borosil make glassware's were used throughout the investigation. These were cleaned with detergent powder followed by rinsing in tap and distilled water. Glasswares were then sterilized in hot air oven at 180°C for 2 hrs. The inoculation needle and other metallic instruments were sterilized by dipping them in alcohol and heating over flame. All precautionary measures were applied to sterilize the inoculation chamber.

The culture media were sterilized in an autoclave at 1.05 kg pressure per square inch (15 lbs psi) for 20 min at 121.6°C. The soil and sand were sterilized at 2.10 kg pressure per square inch (15 lbs psi) for two hours for two consecutive days. The roots of diseased plants were thoroughly washed with tap water and cut into pieces. Root pieces of samples were surface sterilized by dipping them in 1:1000 mercuric chloride solution for 1-2 minutes followed by three washing in sterilized water.

The petriplates containing the media were stored for 24 hrs before use to avoid the possibility of contamination. After isolation cultures of were maintained on PDA medium for further investigations.

3.2 Test pathogen

Sclerotium rolfsii, the causal organism of chickpea collar rot

Fusarium oxysporum f.sp ciceri, the causal organism of chickpea vascular wilt

Rhizoctonia bataticola, the causal organism of chickpea dry root rot

3.3 Culture Media

The following media were used during the course of investigation and the compositions of media are given below.

(i) Potato Dextrose Agar medium (PDA)

Potato (Peeled and sliced)	-	200 g
Dextrose	-	20 g
Agar agar	-	20 g
Distilled water	-	1000 ml

(ii) Potato Dextrose Broth (PDB)

Peeled potato	-	200g
Dextrose	-	20g
Water	-	1000 ml

3.3.1 Potato Dextrose Agar medium

For potato extraction, peeled and sliced potato was taken and boiled in 500 ml water. Then it was strained through muslin cloth. Agar-agar was washed with water and taken in 500 ml of water then heated until it melted fully. Melted agar agar was poured in potato extract and dextrose was added and the solution was finally makeup to one liter. The media was poured in conical flask and sterilized in an autoclave.

3.3.2 Potato Dextrose Broth

Same procedure as described above was employed only agar agar was not added in potato dextrose broth.

3.4 Chemicals

Mercuric chloride solution (1:1000) was prepared and used for surface sterilization of sample during isolation of pathogen and sowing of seed.

C. Method

3.5 Collection, isolation, purification and identification of pathogen

3.5.1 Collection of diseased plant/root sample

Infected plants which showing characteristic symptoms of collar rot (*Sclerotium rolfsii*) wilt (*Fusarium oxysporum* f.sp. *ciceri*) and dry root rot (*Rhizoctonia bataticola*) were collected during the month of October to December 2011 and January 2012 from chickpea fields, JNKVV, Jabalpur. Samples were brought in to the chickpea pathology laboratory for isolation and further studies.

3.5.2 Isolation of plant sample

The diseased parts of root were cut with the help of sharp razor and rinsed with sterilized water to remove the dirt. The pieces were surface sterilized by dipping in mercuric chloride (1:1000) for 1 min followed by three changes of sterilized water and dried in sterilized blotter paper the then transferred aseptically on PDA plates. The plates were incubated at $25\pm 1^{\circ}\text{C}$. The observations were made from the developing colonies. The fungus was sub cultured, purified, identified and maintained on PDA slants.

3.5.3 Identification and purification

Isolation from infected chickpea plants showing symptoms of collar rot, vascular wilt and dry root rot and were made on potato dextrose agar medium. *Sclerotium rolfsii*, *Fusarium oxysporum* f. sp. *ciceri* and *Rhizoctonia bataticola* was found to be associated with root of the diseased plants. The pathogen was identified on the basis of morphological characters with the help of microscope. The purified cultures were maintained on PDA slants.

The cultures of *Sclerotium rolfsii*, *Fusarium oxysporum* f. sp. *ciceri* and *Rhizoctonia bataticola* were purified by sub culturing it through sclerotia single spore and hyphal tip techniques respectively and maintained by mass transfer on PDA slants at $25\pm 2^{\circ}\text{C}$ for further study.

3.5.4 Pathogenicity test

Pot culture technique was employed for proving the pathogenicity test. Mass culture of the fungus was prepared on the chickpea straw inoculated with seven days old culture *Sclerotium rolfsii* and *Fusarium oxysporum* and incubated at $25\pm 1^{\circ}\text{C}$ for 15 days. The inoculum was thoroughly mixed in sterilized soil (sand + soil 1:1) @ 25 g/kg soil. The pots filled with inoculated soil were kept at pot house. Ten seeds of susceptible (JG 62) and resistant (JG 315) varieties were sown in pots along with uninoculated control in three replications. The observations were recorded on per cent mortality due to wilt and the associated pathogen was isolated. However, for dry root rot (*Rhizoctonia bataticola*) Blotter paper technique was employed (Nene *et al.*, 1981).

3.5.5 Screening of chickpea genotypes against collar rot in Net house

The mass culturing of the pathogen was done on sterilized chickpea straw inoculated with culture of *S. rolfsii* and incubated at room temperature ($25\pm 2^{\circ}\text{C}$) for 7 days to obtain the profuse and dense growth of fungal mycelium and sclerotia. The inoculum was thoroughly mixed in sterilized sand + soil (1:1) @ 100 g/ 2 kg soil. The inoculated soil was filled in trays and incubated for 5 days in pot house.

Ten chickpea seeds were sown for each genotype. The pots were watered from time to time and observations were recorded on emergence plant count, percent of seed rot and seedling mortality due to collar rot (up to 30 DAS).

3.5.6 Screening of chickpea genotypes against dry root rot through Blotter paper technique (in vitro)

Blotter paper technique was employed (Nene *et al.*, 1981). For this purpose 5 mm disc of the culture was placed on PDA poured petriplates and incubated at 25°C for 4 days. Five mm disc was cut from the above culture and transferred to 250 ml flasks each containing 100 ml P.D. Broth. After 5 days of incubation at 25°C two mycelial mats were removed from the flask which were added to 100 ml sterilized distilled water in a beaker after its proper crushing for 1 minute in the blender. Surface sterilized seeds of different chickpea genotypes were sown on plastic trays containing sterilized soil + sand (1:1). Five days old seedlings of each genotype were uprooted in such a way so that root system is not disturbed. The root system of these seedlings was properly washed in running water followed by rinsing in sterilized distilled water. The roots of all genotypes (test lines) were dipped in the inoculum kept in a beaker with an up and down movement for about 30 seconds and the excess inoculum was removed by touching the roots to the edge of beaker.

Ten seedlings of the each test line were taken and 10 of these were kept separately on two blotter papers (size 45 cm x 25 cm with one fold). The blotter paper was moistened adequately and the seedlings were kept in such a way so that only cotyledons and roots are covered and the green tops of the seedlings remains outside the blotter paper. The seedlings of a susceptible check BG 212 were also inoculated with each batch of test seedlings.

The folded blotter papers were kept one on top of the other, in heaps of ten in a tray. A susceptible check was kept in each heap. These trays were placed into the incubator at 35°C for 8 days. Artificial light was provided for 12 hrs and the blotter papers were moistened adequately on alternate day.

Rating scale

In order to find out the resistant genotype of chickpea, test lines were scored at the end of the incubation period by examining the seedlings for the extent of root damage and were scored for the disease on 1-9 point scale as mentioned below:

Rating scale used for disease scoring

Rating	Category	Symptoms of DRR	DRR%
1	Resistant	No infection on roots	0.0-10
2-3	Moderately resistant	Very few small lesions on roots	10.1-20
4-5	Tolerant	Lesions on roots clear but small, new roots free from infection	20.1-30
6-7	Susceptible	Lesions on roots many, new roots generally free from lesions	30.1-40
8-9	Highly susceptible	Roots infected and completely discolored	40.1 and above

3.5.7 Screening of chickpea genotypes against wilt and dry root rot resistance under multiple disease sick fields

In order to identify the sources of resistance the chickpea lines were screened in disease sick plot which was prepared by adding the culture of *F. oxysporum* f. sp. *ciceri* in chickpea straw and sand medium. The culture was mixed in the soil (1:9) then added in each row before sowing.

The experiment was conducted against *Fusarium* wilt resistance in randomized block design (RBD) in 2 replications under wilt sick field. For each test line, 40 seeds entry of different trials i.e. Advanced varietal trial (Desi), Initial varietal trial (Desi), Advanced varietal trial (Late sown), Initial varietal trial (Late sown), Initial varietal trial (Rainfed), Advanced varietal trial (Kabuli), Initial varietal trial (Kabuli), Initial varietal trial for extra large seeded Kabuli (EISK), National Nursery Wilt (NNW) were sown in 4 m. long row with seed to seed spacing of 10 cm and row to row spacing of 30 cm.

A susceptible check of JG 62 (early wilter) was maintained after every 2 rows of test entries for comparison (Nene *et al.* 1981). Additionally a wilt resistant cultivar JG 315 was also sown in each trial for comparing the level of resistance in the test entries of the trials. Sowing was done on 20th Nov. 2011 after applying recommended dose of fertilizer. One come up irrigation was given to the crop. Observations on per cent emergence were recorded after

20 days of sowing and mortality due to vascular wilt were recorded weekly up to podding stage. The crop was harvested in the maturity stage (April, 2012)

Percent mortality

Percent mortality was calculated by using the following formula;

$$\text{Percent mortality} = \frac{\text{Number of diseased plants}}{\text{Total number of seedlings}} \times 100$$

Disease reaction – collar rot, vascular wilt and dry root rot

Rating	Category	Reaction type	Percent mortality (%)
1	R	Resistant	0-10
2	MR	Moderately resistant	10.1-20
3	TR	Tolerant	20.1-40
4	S	Susceptible	40.1-60
5	HS	Highly susceptible	60.0 and above

RESULTS

4.1 Collection of diseased samples

Chickpea plants showing typical symptoms of collar rot, wilt and dry root rot were collected at seedling, flowering and pod formation stage from the chickpea pathology field under AICRP on chickpea, (Lead Centre), Seed Breeding Farm, JNKVV, Jabalpur.

4.2 Symptomatology

4.2.1 Collar rot

The disease was found to occur in 15 to 40 days old seedlings. Infected seedlings showed yellowing and can be pulled very easily. *Sclerotium rolfsii* was found to be associated with the collar portion of chickpea plants. The fungus *Sclerotium rolfsii* produces abundant white, fluffy, branched mycelium and numerous sclerotia usually around root/food zone of plant. (Plate 1)

4.2.2 Vascular wilt

The disease is caused by *Fusarium oxysporum* f. sp. *ciceri*. The symptoms observed on wilted chickpea plants exhibited drooping of entire plant parts in the farmers and multiple disease sick field at seedling to flowering and pod formation stage. Infected plants did not showed any external rotting. However, on splitting vertically from collar region downwards showed brown to black discoloration of internal tissues. In adult stage the affected plants showed typical wilting i.e. dropping of leaflets, rachis and lamina. (Plate 2)

4.2.3 Dry root rot

Rhizoctonia bataticola causes the dry root rot disease which occurred at flowering and pod formation stage when the temperature was >30°C. The first symptoms of the disease observed in the field appeared as yellowing of the leaves, within a day or two such leaves dropped and in the course of the next two or three days whole plants dies off. The leaves and

stem of the affected plants exhibited straw coloured appearance. The tap root becomes dark and quite brittle in dry soil. The dark black sclerotia were observed within host tissues when the stem of diseased plant is examined. On uprooting of the plants from the soil, the basal stem and main root shows dry rot symptoms. The tissues are weakened and break off easily. (Plate 3)

4.3 Isolation, purification and identification of pathogens

Isolation from infected chickpea plants showing symptoms of collar rot, wilt and dry root rot, were made on potato dextrose agar medium. The pathogens viz., *S. rolfsii*, *F. oxysporum* f.sp. *ciceri* and *R. bataticola* was found to be associated with root of all samples.

The pathogens were identified on the basis of cultural and morphological characters with the help of microscope; purified by single hyphal tip and single spore technique and maintained by mass transfer on PDA slants at 25°C for further investigation.

4.4 Pathogenicity test

Pathogenicity test for the three pathogens causing collar rot, vascular wilt and dry root rot of chickpea was conducted by using the susceptible check JG 62 (wilt and root rot), BG 212 (dry root rot) and resistant check JG 315 and JG 63 through pot inoculation and blotter paper techniques.

4.4.1 Vascular wilt

Pathogenicity test for the pathogen causing vascular wilt was conducted by using susceptible JG 62 and JG 315 as resistant varieties through pot inoculation technique. The test pathogen *F. oxysporum* was found pathogenic causing cent per cent wilting within 20 days in inoculated JG 62 plants. However, no wilting was observed in resistant variety JG 315.

4.4.2 Collar rot

Pathogenecity test conducted against collar rot under artificially inoculated condition in Net house revealed that the disease occurred within 15 days after sowing. Affected seedlings turn yellow, which can be easily pulled out. White mycelial strands near the collar region of the infected plant were observed.

4.4.3 Dry root rot

Pathogenecity test was conducted by using susceptible variety BG 212 through blotter paper techniques (Nene *et al.*, 1981). Plant showed clear cut symptoms of dry root rot. Primary and the secondary root become black, distorted and brittle. On reisolation from the root of infected plants exhibited the growth of *R. bataticola* on PDA.

4.5 Succession of soil borne diseases of chickpea

To know the succession of soil borne diseases of chickpea occurring from seedling to flowering and pod formation stages, weekly observations on plant mortality due to collar rot (*Sclerotium rolfsii*), vascular wilt (*Fusarium oxysporum*) and dry root rot (*Rhizoctonia bataticola*) were recorded and correlated with the weather data. (Nov. 2011 – March 2012)

The progressive development of soil borne disease of chickpea was investigated in relation in environment factors by using a extra large seeded kabuli chickpea genotype Kripa (Phule G 0517) and small seed desi chickpea (CSJ 515) grown in multiple disease sick field alongwith susceptible checks JG 62 and BG 212 (collar rot, wilt and dry root rot) (Table and graph)

4.5.1 Collar rot and vascular wilt

The data presented in the Table 1 indicated that maximum collar rot 28.2 per cent and 22.5 percent vascular wilt in desi variety (CSJ 515) and 30.5 per cent collar rot and 25.2 vascular wilt in kabuli variety Kripa (Phule G 0517) was observed in the 52nd standard week i.e. 5th week of Dec.2011. During that period the temperature ranged from 7.4⁰C to 24.9⁰C and relative humidity 24.0 to 89 per cent.

4.5.2 Dry root rot

Increased percent mortality due to dry root rot i.e. 25.3 percent in desi variety CSJ 515 and 29.2 percent in Kabuli Kripa (Phule G- 0577) was observed in the 10th standard week i.e. 2nd week of March 2012. When the maximum temperature was 30.8⁰C ranging from 11.6-30.8⁰C coupled with the range of relative humidity 18 to 76 per cent. However earlier the incidence was 17.2 percent in desi variety CSJ 515 and 15.4 per cent in Kabuli variety Kripa (Phule G 0517) was observed in the 7th standard week. Which

increased dramatically from 15.4 to 29.2 per cent in desi and 17.2 to 25.3 per cent in kabuli during 10th week.(March 2012). No disease was observed in 51st, 52nd, 1st and 6th standard week when the maximum temperature ranged from 20.8 to 26.4⁰ C. But the disease started after that when the maximum temperature rises with decrease in relative humidity. Succession of disease is positively correlated with the increase /decrease in temperature as well as relative humidity.

Table 1 : Progressive development of collar rot, vascular wilt and dry root rot observed under field condition on Desi (CSJ 515 Genotype)

Month	Week	S.D. Week	Collar rot	Wilt	DRR	Temperature		Rainfall (mm)	Relative Humidity	
						Max.	Min.		Max.	Min.
December 2011	Iv	51	27.2	20.2	0.0	24.8	5.3	00	90	24
	V	52	28.5	22.5	0.0	24.9	7.4	00	89	24
January 2012	I	01	25.2	15.36	0.0	23	12.2	28.6	95	69
	Ii	02	17.5	13.1	0.0	20.2	6.3	00	93	41
	Iii	03	0.0	18.2	0.0	23.7	7.1	13.2	90	36
	Iv	04	0.0	0.0	0.0	21.5	9.0	7.6	89	57
February	I	05	0.0	0.0	0.0	20.8	8.0	00	91	46
	Ii	06	0.0	0.0	0.0	26.4	8.8	00	89	57
	Iii	07	0.0	0.0	17.2	27.6	10.4	00	90	42
	Iv	08	0.0	0.0	23.5	30.5	9.4	00	86	28
March	I	09	0.0	0.0	22.4	30.1	10.4	00	86	24
	Ii	10	0.0	0.0	25.3	30.8	11.6	00	76	18

Table 2. Progressive development of collar rot, vascular, wilt and dry root rot observed on under field condition Kabuli Kripa (Phule G 0517) Genotype

Month	Week	S.D. Week	Collar rot	Wilt	DRR	Temperature		Rainfall (mm)	Relative Humidity	
						Max.	Min.		Max.	Min.
December 2011	iv	51	28.4	23.4	0.0	24.8	5.3	00	90	24
	v	52	30.5	25.2	0.0	24.9	7.4	00	89	24
January 2012	i	01	25.2	19.4	0.0	23.0	12.2	28.6	95	69
	ii	02	15.1	11.2	0.0	20.2	6.3	00	93	41
	iii	03	0.0	10.0	0.0	23.7	7.1	13.2	90	36
	iv	04	0.0	0.0	0.0	21.5	9.0	7.6	89	57
February	i	05	0.0	0.0	0.0	20.8	8.0	00	91	46
	ii	06	0.0	0.0	0.0	26.4	8.8	00	89	57
	iii	07	0.0	0.0	15.4	27.6	10.4	00	90	42
	iv	08	0.0	0.0	27.5	30.5	9.4	00	86	28
March	i	09	0.0	0.0	25.4	30.1	10.4	00	86	24
	ii	10	0.0	0.0	29.2	30.8	11.6	00	76	18

4.5.3 Correlation studies

(I) Collar rot- Correlation studies revealed that minimum temperature, minimum relative humidity and rainfall exhibited Positive correlation $r=0.0817$, -0.391 and -0.338 in kabuli variety and 0.1169 , 0.0812 and -0.299 in desi variety. Further maximum Temperature, maximum Relative humidity showed negative correlation $r=-0.9885$, 0.582 , kabuli variety and 0.9848 , 0 -0.546 desi variety respectively with collar rot disease in chickpea.

(ii) Vascular wilt- Correlation studies revealed that minimum temperature, minimum relative humidity and rainfall exhibited Positive correlation $r = -0.0011$, -0.414 , -0.043 in kabuli variety and -0.298 , -0.693 , -0.361 desi variety. Further maximum Temperature, rainfall and maximum Relative humidity showed negative correlation $r = 0.9947$, -0.043 , -0.646 in kabuli and 0.9252 , -0.361 , and -0.842 in desi variety respectively with vascular wilt disease in chickpea.

(iii) Dry root rot-

Correlation studies revealed that minimum temperature, and minimum relative humidity exhibited Positive correlation $r = -0.756, 0.8001$ in kabuli variety and $0.2598, -0.316$ in desi variety. Further maximum Temperature, maximum relative humidity showed negative correlation $r = 0.3851, 0.1271$ in kabuli variety and $0.9866, 0. -0.828$ desi variety respectively with dry root rot disease in chickpea.

Table;03 Correlation studies on collar rot ,vascular wilt and dry root rot

Kabuli Kripa (Phule G 0517)		Correlation value	Desi CSJ 515		Correlation value
Collor rot + Temp max.		0.9885	Collor rot + Temp max.		0.9848
	min	0.0817		min	0.1169
	rainfall	0.0391		rainfall	0.0812
	RH max	-0.582		RH max	-0.546
	min	-0.338		min	-0.299
Wilt + Temp. max		0.9947	Wilt + Temp. max		0.9252
	min	0.0011		min	-0.298
	rainfall	-0.043		rainfall	-0.361
	RH max	-0.646		RH max	-0.842
	min	-0.414		min	-0.693
Dry root +	Temp. max.	0.3851	Dry root +	Temp. max.	0.9866
	min	-0.756		min	0.2598
	rainfall	0		rainfall	0
	RH max	0.1271		RH max	0.2098
	min	0.8001		min	-0.316

*Correlation is significant at the 0.05 level (2 tailed)

**Correlation is significant at the 0.01level (2 tailed)

5.1 Host resistance

To identify the sources of host resistance against collar rot, vascular wilt and dry root rot pathogen, 159 lines comprising of 98 desi and 61 kabuli chickpea lines were screened alongwith resistant JG 315 and susceptible check JG 62 in artificially inoculated condition in Net house, Sick field and Blotter paper techniques, respectively. Each entry was sown in two replications. Observation on emergence count and plant mortality due to collar rot, vascular wilt and dry root rot were recorded after 20 days of sowing and at 7 days interval up to pod filling stage respectively. The lines exhibiting <10 per cent wilting were considered as resistant, 10 – 20 per cent as moderately resistant, 20.1- 30 per cent as tolerant, 30.1 to 40 per cent as considered as susceptible and >40 were considered highly susceptible. The results of screening of the chickpea lines against wilt are being summarized. (Tables – 4-11.)

5.1.1 Host plant resistance against soil borne diseases of chickpea

To know the individual and multiple disease resistance for collar rot, vascular wilt and dry root rot 159 entries comprising of 98 desi and 61 kabuli type were evaluated under artificial inoculated condition and in wilt sick field .

5.1.2 Collar rot (desi genotype)

The data presented in table 4 indicate that among the 98 entries desi type screened for collar rot ,15 entries namely GJG 0809, GNG 1958, GL 27104, IPS 06127, PG 0105, GNG 2065, Phule G 0204 , IPC 07-19 , H-08-13 ,GNG 2068, IPC 07-09, GJG 0907, GL 27091, IPC 06- 127,GNG 1581.were found resistant and 16 entries as moderately resistant i.e. Phule G 07102, H 07163, GL 26054, GNG 1999, GJG 092,GNG-0921,JG 24, Phule G 02152, GJG 0907, GSJ 313, RVSSG 10, GJG 0809, GJG 0814, H 08 75, CSJ 515 BG 3023 NDG 1105 to collar rot. Seedling mortality due to collar rot ranged from 0.0 to 89.9 per cent and seed rot ranged from 0 to 40 percent while in resistant check (JG 315) it was 12.5 per cent in susceptible check (JG 62) 78.4 per cent mortality was observed.

Table 4 : Screening of chickpea desi entries against seed rot and seedling mortality due to *Sclerotium rolfsii* under artificially inoculated pot condition

S.No.	En. Code	En. name	Total plants		Seed rot (%)		Seedling mortality (%)		Average (%)	
			RI	RII	RI	RII	RI	RII	Seedling mortality	Seed rot
1	P ₁	NBeG 49	5.	4	0.0	20.0	60.0	75.0	67.5	10.0
2	P ₂	GJG 0809	5.	5	0.0	0.0	0.0	0.0	0.0	0.0
3	P ₃	Phule G07102	5.	4	0.0	20.0	20.0	15.0	17.5	10.0
4	P ₄	BG3013	4.	5	20.0	0.0	75.0	80.0	77.5	10.0
5	P ₅	PBG 5	5	5	0.0	0.0	40.0	40.0	40.5	0.0
6	P ₆	GNG1958(AVT-2)	5	4	0.0	20.0	0.0	0.0	0.0	10.0
7	P ₇	GL27104	4	5	20.0	0.0	0.0	0.0	0.0	10.0
8	P ₈	GNG1581	3	4	40.0	20.0	0.0	0.0	0.0	30.0
9	P ₉	GNG 1936	4	5	20.0	0.0	40.0	50.0	45.0	10.0
10	P ₁₀	GL27091	4	4	20.0	20.0	0.0	0.0	0.0	20.0
11	P ₁₁	H 07-163	4	5	20.0	0.0	15.0	20.0	17.5	10.0
12	P ₁₂	GNG 469	4	4	20.0	20.0	33.3	30.3	31.5	20.0
13	P ₁₃	GL 26054	4	5	20.0	00.0	20.0	15.0	17.5	10.0
14	P ₁₄	GNG 1999	5	4	0.0	20.0	10.0	15.0	12.5	10.0
15	P ₅₁	BG 3031	4	3	20.0	40.0	25.0	20.0	22.5	30.0
16	P ₅₂	Phule G 0204-16	5	5	0.0	0.0	20.0	20.0	20.0	0.0
17	P ₅₃	GJG 0922	5	5	0.0	0.0	40.0	40.0	40.0	0.0
18	P ₅₄	BGD 1071	4	4	20.0	20.0	40.0	25.0	32.5	20.0
19	P ₅₅	CSJ 697	5	5	0.0	0.0	20.0	20.0	20.0	0.0

20	P ₅₆	H 08-25	5	5	0.0	0.0	60.0	70.0	65.0	0.0
21	P ₅₇	GNG 0921	3	3	40.0	40.0	20.0	20.0	20.0	40.0
22	P ₅₈	Phule G06102	5	5	0.0	0.0	80.0	80.0	80.0	0.0
23	P ₅₉	IPC 06-127	5	5	0.0	0.0	10.0	10.0	10.0	0.0
24	P ₆₀	JG 28	4	5	20.0	0.0	20.0	25.0	22.5	10.0
25	P ₆₁	NDG 1105	5	5	0.0	0.0	10.0	10.0	10.5	0.0
26	P ₆₂	GNG2066	5	5	0.0	0.0	20.0	40.0	30.0	0.0
27	P ₆₃	JG 27	5	5	0.0	0.0	20.0	40.0	30.0	0.0
28	P ₆₄	IPC 07-56	5	5	0.0	0.0	50.0	40.0	45.0	0.0
29	P ₆₅	PG 0101	5	5	0.0	0.0	20.0	40.0	30.0	0.0
30	P ₆₆	RVSSG	5	5	0.0	0.0	20.0	20.0	20.0	0.0
31	P ₆₇	IPC 2006-84	3	3	40.0	40.0	33.3	40.0	36.6	40.0
32	P ₆₈	JG 14	4	5	20.0	0.0	50.0	40.0	45.0	10.0
33	P ₆₉	BGD 1072	5	5	0.0	0.0	20.0	20.0	20.0	0.0
34	P ₇₀	IPC 2006-77	4	5	20.0	0.0	40.0	25.0	32.5	10.0
35	P ₇₁	RSG 963	5	5	0.0	0.0	10.0	10.0	10.5	0.0
36	P ₇₂	H 08-75	5	5	0.0	0.0	60.0	70.0	65.0	0.0
37	P ₇₃	CSJ753	5	5	0.0	0.0	40.0	25.0	32.5	0.0
38	P ₇₄	BG 3031	5	5	0.0	0.0	40.0	40.0	40.0	0.0
39	P ₇₅	GNG 2081	4	5	20.0	0.0	50.0	40.0	45.0	10.0
40	P ₂₁	BG 3024	4	5	20.0	0.0	25.0	20.0	22.5	10.0
41	p ₂₂	BDNG 2010-1	3	4	40.0	20.0	33.3	25.0	29.1	30.0
42	P ₂₃	BGD 1068	3	3	40.0	40.0	33.3	33.3	33.3	40.0
43	P ₂₄	JG 24	4	3	20.0	40.0	20.0	15.0	17.5	20.0
44	P ₂₅	PG 0105	5	5	0.0	0.0	10.0	10.0	10.0	0.0
45	P ₂₆	Phule G0215-2	4	4	20.0	20.0	20.0	20.0	10.0	20.0
46	P ₂₇	RVSSG 9	5	4	0.0	20.0	25.0	20.0	22.5	10.0
47	P ₂₈	GJG 0907	3	3	40.0	40.0	10.0	10.0	10.0	40.0
48	P ₂₉	GSJ 313	3	4	40.0	20.0	10.0	20.0	15.0	30.0
49	P ₃₀	RVSSG 10	5	5	0.0	0.0	20.0	20.0	20.0	0.0
50	P ₃₁	GNG 2065	4	5	20.0	0.0	10.0	10.0	10.0	10.0
51	P ₃₂	Phule G 0204-4	5	5	0.0	0.0	10.0	10.0	10.0	0.0

52	P ₃₃	GJG 0906	4	3	20.0	40.0	50.0	33.3	41.6	30.0
53	P ₃₄	CSJ 0564	5	5	0.0	0.0	40.0	40.0	40.0	0.0
54	P ₃₅	GNG 2064	5	5	0.0	0.0	40.0	40.0	40.0	0.0
55	P ₃₆	GJG0809	5	5	0.0	0.0	20.0	20.0	20.0	0.0
56	P ₃₇	BG 3023	5	4	0.0	20.0	10.0	10.10	10.5	10.0
57	P ₃₈	CSJ303	4	5	20.0	0.0	25.0	20.0	22.5	10.0
58	P ₃₉	AKG 1001	5	5	0.0	0.0	40.0	25.0	32.5	0.0
59	P ₄₀	IPC 07-19	5	5	0.0	0.0	10.0	10.1	10.0	0.0
60	P ₄₁	H 08-13	5	5	0.0	0.0	10.0	10.0	10.0	0.0
61	P ₄₂	BG 3004	4	5	20.0	0.0	25.0	20.0	22.5	10.0
62	P ₄₃	IPC 08-11	4	5	20.0	0.0	40.0	40.0	40.0	10.0
63	P ₄₄	H 08-11	4	5	20.0	0.0	20.0	20.0	20.0	10.0
64	P ₄₅	JG 23	4	3	20.0	40.0	50.0	33.3	41.6	30.0
65	P ₄₆	GL 23094	5	5	0.0	0.0	20.0	20.0	20.0	0.0
66	P ₄₇	H 04-99	5	5	0.0	0.0	40.0	25.0	32.5	0.0
67	P ₄₈	GL 26069	5	5	0.0	0.0	40.0	40.0	40.0	0.0
68	P ₉₁	BGD 1070	4	4	20.0	20.0	80.0	80.2	81.0	20.0
69	P ₉₂	GJG 0910	5	5	0.0	0.0	50.0	45.0	47.5	0.0
70	P ₉₃	Vijay	4	3	20.0	40.0	80.5	83.3	81.9	30.0
71	P ₉₄	BGM 572	4	4	20.0	20.0	75.0	67.5	70.2	20.0
72	P ₉₅	BCP 60	5	5	0.0	0.0	80.2	81.2	80.7	0.0
73	P ₉₆	CSJ 730	5	5	0.0	0.0	25.2	22.5	23.8	0.0
74	P ₉₇	JG 25	5	5	0.0	0.0	69.5	64.7	67.1	0.0
75	P ₉₈	GJG0814	5	5	0.0	0.0	20.0	20.0	20.0	0.0
76	P ₉₉	CSJ 513	5	5	0.0	0.0	80.2	82.7	81.4	0.0
77	P ₁₀₀	JG 26	5.	5	0.0	0.0	79.2	77.2	78.2	0.0
78	P ₁₀₁	H 08-93	5	5	0.0	0.0	68.2	68.8	68.5	0.0
79	P ₁₀₂	BG 3030	4	5	20.0	0.0	60.3	62.8	62.5	10.0
80	P ₁₀₃	GJG 0904	4	5	20.0	0.0	72.4	72.7	72.5	10.0
81	P ₁₀₄	BGD 1069	4	5	20.0	0.0	60.5	61.9	61.2	10.0
82	P ₁₀₅	RSG 931	4	5	20.0	0.0	25.0	32.5	28.7	10.0
83	P ₁₀₆	GNG 2002	5	4	0.0	20.0	67.5	67.8	67.2	10.0

84	P ₁₀₇	GJG 0825	5	5	0.0	0.0	35.0	75.4	55.2	0.0
85	P ₁₀₈	H 08-75	5	5	0.0	0.0	20.0	17.5	18.7	0.0
86	P ₁₀₉	IPC 08-68	5	4	0.0	20.0	50.0	45.0	47.5	10.0
87	P ₁₁₀	Phule G 09103	4	5	20.0	0.0	35.0	32.5	33.5	10.0
88	P ₁₁₁	GNG 2068	4	5	20.0	0.0	10.0	10.0	10.0	10.0
89	P ₁₁₂	RSG 888	5	5	0.0	0.0	30.0	32.5	31.2	0.0
90	P ₁₁₃	BG 3029	5	4	0.0	20.0	80.3	82.2	82.2	10.0
91	P ₁₁₄	GNG 2085	5	5	0.0	0.0	90.2	89.8	89.8	0.0
92	P ₁₁₅	IPC 07-09	5	5	0.0	0.0	10.1	10.0	10.0	0.0
93	P ₁₁₆	BGD 1061	4	5	20.0	0.0	60.0	62.5	62.5	10.0
94	P ₈₁	RSG 888	5	5	0.0	0.0	80.0	75.0	77.5	0.0
95	P ₈₂	CSJ 515	5	5	0.0	0.0	20.0	20.0	20.0	0.0
96	P ₈₃	RSG 931	5	5	0.0	0.0	25.0	20.0	22.5	0.0
97	P ₈₄	GNG 1996	5	5	0.0	0.0	80.0	80.0	80.0	0.0
98	P ₈₅	BGM572	4	5	20.0	0.0	50.0	60.0	55.0	10.0
99	JG 315 (R) check		4	5	0.0	0.0	10.0	15.0	12.5	10.0
100	JG62(S) check		4	5	20.0	20.0	79.9	77.0	78.4	20.0
	L.S.I.								35.8	

R= Resistant, S=Susceptible

5.1.2 Collar rot (kabuli genotypes)

The data presented in table 5 indicate that among the 61 entries (kabuli) evaluated out of which 15 entries namely CSJK 68, HK 06 -171, phuleG-09311,IPCK 08 120, IPCK 06-78 , CSJK-54, GLK 26162, H K 06 152, Phul e G 09316, HK 08-212, CSJK 74, IPS-06-163 ,RVSSG 11, GNG 2112 , IPS-06-143 were found resistant and 7 entries *i.e.*(HK 08- 231, BG 3025, GNG 2104, Phule G 04305, CSJK 6, IPCK 06-56, GNG 1969) as moderately resistant. The collar rot disease incidence ranged from 10.0 to 94.7, per cent and seed rot range from 0.0 to 30 per cent. Resistant check (JG 315) mortality 11.0 percent and susceptible check (JG 62) mortality 61.2 per cent were recorded.

Table 5. Screening of chickpea kabuli entries against seed rot and seedling mortality due to *Sclerotium rolfsii* Under artificially inoculated pot condition.

S. No.	En. Code	En. name	Total plants		Seed rot (%)		Seedling mortality (%)		Average (%)	
			RI	RII	RI	RII	RI	RII	Collar rot	Seed rot
1	P ₁₇₅	Phule G 09305	4	5	20.0	0.0	60.0	67.0	63.50	10.0
2	P ₁₇₆	JGK 17	5	5	0.0	0.0	71.10	75.0	73.00	0.0
3	P ₁₇₇	IPCK 08-136	4	4	20.0	0.0	30.4	40.5	35.40	20.0
4	P ₁₇₈	BG 3012	5	5	0.0	0.0	82.3	85.4	83.80	0.0
5	P ₁₇₉	HK 08-206	5	5	0.0	0.0	40.0	85.5	37.6	0.0
6	P ₁₈₀	JGK 13	5	5	0.0	0.0	60.0	62.3	61.1	0.0
7	P ₁₈₁	Vihar (Phule G 95311)	4	5	20.0	0.0	84.5	82.3	83.4	10.0
8	P ₁₈₂	IPCK 08-120	5	5	0.0	0.0	10.0	10.0	10.0	0.0
9	P ₁₈₃	JGK 18	5	4	0.0	20.0	30.3	33.3	31.8	10.0
10	P ₁₈₄	AKG 2002-1K	5	5	0.0	0.0	35.0	40.0	37.5	0.0
11	P ₁₈₅	CSJK 72	5	5	0.0	0.0	83.2	84.2	83.7	0.0
12	P ₁₈₆	JGK 16	4	5	20.0	0.0	87.5	89.5	88.5	10.0
13	P ₁₈₇	IPCK 08-130	4	4	20.0	20.0	94.2	95.2	94.7	20.0
14	P ₁₈₈	CSJK 70	5	5	0.0	0.0	80.5	86.7	83.6	0.0
15	P ₁₈₉	Kripa (Phule G0517)	4	5	20.0	0.0	83.2	84.2	83.7	10.0
16	P ₁₉₀	CSJK 66	4	4	20.0	20.0	91.2	94.4	92.8	20.0
17	P ₁₉₁	JGK 19	5	5	0.0	0.0	60.5	63.3	61.9	0.0
18	P ₁₉₂	BG 3027	5	4	0.0	20.0	66.2	69.3	67.7	10.0
19	P ₁₅₁	Vihar	4	3	20.0	40.0	89.2	88.5	88.8	30.0
20	P ₁₅₂	HK 08-231	4	4	20.0	20.0	20.0	10.1	15.1	20.0
21	P ₁₅₃	CSJK68	5.	5	0.0	0.0	10.0	10.0	10.0	0.0
22	P ₁₅₄	BG3025	3	4	40.0	20.0	20.0	10.0	15.0	30.0
23	P ₁₅₅	RVSSG11	5	5	0.0	0.0	10.0	10.0	10.0	0.0

24	P ₁₅₆	GNC2112	5	5	0.0	0.0	10.0	10.0	10.0	0.0
25	P ₁₅₇	HK 06-171	5	5	0.0	0.0	10.0	10.0	10.0	0.0
26	P ₁₅₈	IPCK 07-62	4	5	20.0	0.0	20.0	25.0	22.5	10.0
27	P ₁₅₉	RVSSG 12	3	4	40.0	20.0	20.0	25.0	22.5	30.0
28	P ₁₆₀	Phule G 09311	5	5	0.0	0.0	10.0	10.0	10.0	0.0
29	P ₁₆₁	CSJK 54	5	5	0.0	0.0	10.0	10.0	10.0	0.0
30	P ₁₆₂	Phule G 09316	5	5	0.0	0.0	10.0	10.0	10.0	0.0
31	P ₁₆₃	GNG 2104	5	5	0.0	0.0	20.0	10.0	15.0	0.0
32	P ₁₆₄	HK 08-212	5	5	0.0	0.0	10.1	10.0	10.0	0.0
33	P ₁₆₅	CSJK 74	5	5	0.0	0.0	10.0	10.0	10.0	0.0
34	P ₁₆₆	GNG 2047	4	5	20.0	0.0	25.0	20.0	22.5	10.0
35	P ₁₆₇	IPC 06-143	5	5	0.0	0.0	10.0	10.0	10.0	0.0
36	P ₁₆₈	HK 06-163	5	5	0.0	0.0	10.0	10.0	10.0	0.0
37	P ₁₂₁	JGK 2005-301	4	4	20.0	20.0	80.0	75.0	77.5	20.0
38	P ₁₂₂	CSJK 66	5	5	0.0	0.0	60.0	50.0	55.0	0.0
39	P ₁₂₃	Phule G 0027	4	4	20.0	20.0	75.0	75.0	75.0	20.0
40	P ₁₂₄	JGK 2003-04	4	5	20.0	0.0	89.0	90.0	89.5	10.0
41	P ₁₂₅	CSJK-1	5	4	0.0	20.0	90.0	95.0	92.5	10.0
42	P ₁₂₆	GNG 1969	3	4	40.0	20.0	66.0	60.0	63.3	30.0
43	P ₁₂₇	GLK 26155	4	3	20.0	40.0	90.0	92.3	91.1	30.0
44	P ₁₃₀	SC-RS-1	4	5	20.0	0.0	25.0	40.0	32.5	10.0
45	P ₁₃₁	HK 94-134	5	5	0.0	0.0	80.0	85.0	82.5	0.0
46	P ₁₃₂	GNG 1888	4	5	20.0	0.0	89.0	91.5	90.2	10.0
47	P ₁₃₃	IPCK 113	5	4	0.0	20.0	75.0	67.0	71.0	10.0
48	P ₁₃₄	JGK 1	4	5	20.0	0.0	75.0	80.0	77.5	10.0
49	P ₁₃₅	HK 07-234	4	3	20.0	40.0	50.0	40.0	45.0	30.0
50	P ₁₃₆	Phule G 04305	5	4	0.0	20.0	20.0	20.0	20.0	10.0
51	P ₁₃₇	CSJK 6	5	5	0.0	0.0	10.0	20.0	15.0	0.0
52	P ₁₃₈	IPCK 06-78	5	5	0.0	0.0	10.0	10.0	10.0	0.0
53	P ₁₃₉	CSJK 42R	4	4	20.0	20.0	80.0	85.0	82.5	20.0
54	P ₁₄₀	GNG 2034	4	5	20.0	0.0	50.0	40.0	45.0	10.0
55	P ₁₄₁	GLK 26162	5	5	0.0	0.0	10.0	10.0	10.0	0.0
56	P ₁₄₂	IPCK 06-56	5	5	0.0	0.0	10.0	20.0	15.0	0.0

57	P ₁₄₃	GNG 1969	5	5	0.0	0.0	10.0	20.0	15.0	0.0
58	P ₁₄₄	HK 06-152	5	5	0.0	0.0	10.0	10.0	10.0	0.0
59	P ₁₇₁	GLK 28127	4	5	20.0	0.0	25.0	20.0	22.5	10.0
60	P ₁₇₂	Kripa (Phule G0517)	3	4	40.0	20.0	33.3	50.0	41.6	30.0
61	P ₁₇₃	GLK 26167	4	4	20.0	20.0	50.0	50.0	50.0	20.0
62	JG 315(R) Check		5	5	0.0	0.0	13.5	8.5	11.0	0.0
63	JG62(S) Check		4	5	20.0	0.0	62.0	60.4	61.2	10
	L.S.I								45.3	

R=Resistance, S=Susceptible.

5.2.1 Vascular wilt (desi genotype)

In this trial 98 desi entries were evaluated for wilt resistance among them 9 entries viz, BDNG 2010-1, GNG 2064, IPC 08-11, H 093, BGD 1069, GNG 2002, GNG 2085, BG-3030, H 08-13. were exhibited resistant and 12 entries namely viz JG 24, AKG 1001, BG 30040, JG 25, CSJ 513, H 08-71, BGD 1061, GJG 0922, IPC 07-56, GNG 0921, GNG 0910, BCP-60 were found to be moderately resistant. The disease incidence ranged from 6.5 to 98.2 percent. Hundred percent wilting was observed in susceptible check JG 62 whereas in resistant check (7.5 %). (Table 6)

5.2.1 Dry root rot (desi genotype)

Out of 98 desi entries were screened for dry root rot, among them 48 entries showed resistant reaction and 28 entries as moderately resistant. The disease incidence ranged from 1.9 to 54.5 percentages. The percent mortality due to dry root rot in susceptible check BG 212 has been observed 66.3 percent. The least severity index (LSI) remained below 20 percent hence the entries exhibiting resistant and moderately resistant reaction to the disease were not consider to draw any conclusion on the basis of field data (Table 6).

Table 06; Screening of chickpea desi entries against vascular wilt and dry root rot under sick field condition

S. No.	En. code	En.name	Total plants		Percent mortality (%)				Average (%)	
					Wilt		Dry root rot		Wilt	DRR
			R I	R II	R I	R II	R I	R II	R I	R II
1	P ₉₁	BGD 1070	52	52	98.0	98.0	1.9	2.6	98.0	2.6
2	P ₉₂	GJG 0910	50	52	20.0	19.2	44.0	38.4	19.6	41.2
3	P ₉₃	Vijay	45	40	75.5	83.3	2.2	2.5	79.4	2.3
4	P ₉₄	BGM 572	57	54	84.2	86.2	15.7	14.8	85.2	15.2
5	P ₉₅	BCP 60	44	48	18.1	15.1	13.6	12.5	16.6	13.0
6	P ₉₆	CSJ 730	50	47	76.0	80.2	2.0	2.1	78.1	2.0
7	P ₉₇	JG 25	45	42	13.8	14.2	60.0	57.0	14.0	58.0
8	P ₉₈	GJG 0814	50	45	26.0	20.0	2.0	2.2	23.0	2.1
9	P ₉₉	CSJ 513	45	42	18.0	19.0	88.0	7.1	18.5	7.9
10	P ₁₀₀	JG 26	55	48	69.0	85.4	3.6	6.2	77.0	4.9
11	P ₁₀₁	H 08-93	46	48	10.8	8.3	4.3	4.1	9.3	4.2
12	P ₁₀₂	BG 3030	45	38	4.9	9.5	15.5	10.5	7.2	13.0
13	P ₁₀₃	GJG 0904	30	36	78.2	75.0	6.6	5.5	76.6	6.0
14	P ₁₀₄	BGD 1069	44	38	10.5	5.2	4.5	5.2	7.8	4.8
15	P ₁₀₅	RSG 931	40	45	85.7	80.0	7.8	10.5	82.8	9.1
16	P ₁₀₆	GNG 2002	49	47	4.0	9.4	6.1	6.6	6.7	6.3
17	P ₁₀₇	GJG 0825	47	37	40.5	35.1	12.7	16.2	37.8	14.4
18	P ₁₀₈	H 08-71	45	44	15.5	13.6	8.8	9.0	14.5	8.9
19	P ₁₀₉	IPC 08-68	40	45	26.0	20.0	12.3	15.5	23.0	13.9

20	P ₁₁₀	Phule G 09103	50	45	29.5	22.2	2.0	2.2	25.8	2.1
21	P ₁₁₁	GNG 2068	36	43	94.7	97.6	2.7	2.3	96.1	2.5
22	P ₁₁₂	RSG 888	45	38	46.6	57.7	22.2	26.3	52.1	24.2
23	P ₁₁₃	BG 3029	35	37	60.5	59.7	2.8	2.7	60.0	2.7
24	P ₁₁₄	GNG 2085	34	45	11.7	4.0	2.9	4.4	7.8	3.6
25	P ₁₁₅	IPC 07-09	40	44	24.4	19.0	5.6	6.2	21.7	5.9
26	P ₁₁₆	BGD 1061	45	52	15.5	10.5	50.5	54.5	13.0	54.5
27	P ₂₁	BG 3024	23	24	49.0	45.8	13.0	8.3	47.4	10.6
28	P ₂₂	BDNG 2010-1	43	41	11.6	7.3	4.6	73.0	9.4	5.9
29	P ₂₃	BGD 1068	34	28	94.1	96.4	8.1	8.3	95.2	8.1
30	P ₂₄	JG 24	43	40	13.7	11.1	6.9	7.5	12.4	7.2
31	P ₂₅	PG 0105	40	48	50.2	52.0	10.0	15.5	51.1	12.7
32	P ₂₆	Phule G 0215-2	38	49	32.2	40.0	7.8	8.1	36.1	7.9
33	P ₂₇	RVSSG 9	42	50	83.3	90.0	7.1	4.0	86.6	5.5
34	P ₂₈	GJG 0907	43	40	60.5	68.1	14.2	11.6	64.3	12.9
35	P ₂₉	CSJ 313	50	43	80.0	69.7	18.2	19.5	74.8	18.9
36	P ₃₀	RVSSG 10	44	39	40.9	45.0	22.7	25.6	42.9	24.1
37	P ₃₁	GNG 2065	49	45	40.9	44.8	2.0	22.0	42.8	2.1
38	P ₃₂	Phule G 0204-4	20	23	22.4	26.0	10.0	8.6	24.2	9.3
39	P ₃₃	GJG 0906	44	50	85.0	80.0	18.0	15.2	82.5	16.6
40	P ₃₄	CSJ 0564	49	41	53.0	60.0	4.0	4.8	56.5	4.4
41	P ₃₅	GNG 2064	38	42	10.5	9.5	2.6	3.2	10.0	2.9
42	P ₃₆	GJG 0809	46	59	91.0	86.0	4.3	6.3	88.5	5.3
43	P ₃₇	BG 3023	37	42	90.0	84.0	8.1	7.1	87.0	7.6
44	P ₃₈	CSJ 303	40	34	68.0	70.5	12.5	11.7	69.2	12.5
45	P ₃₉	AKG 1001	38	34	13.1	20.0	2.6	2.9	16.5	2.7
46	P ₄₀	IPC 07-19	51	56	18.0	24.0	7.5	9.2	21.0	8.3
47	P ₄₁	H 08-13	44	49	4.5	12.0	6.4	3.5	8.2	4.9
48	P ₄₂	BG 3004	45	50	8.8	14.0	2.2	2.0	11.4	2.1
49	P ₄₃	IPC 08-11	54	58	9.2	5.1	1.8	2.5	7.1	2.1
50	P ₄₄	H 08-18	24	21	75.0	82.2	25.0	28.5	78.6	26.7
51	P ₄₅	JG 23	47	51	68.0	74.5	3.9	4.2	71.2	4.0
52	P ₁	NBeG 49	34	41	88.2	80.4	8.8	12.1	84.3	10.4
53	P ₂	GJG 0809	43	50	86.0	82.1	11.6	16.0	84.0	13.8

54	P ₃	Phule G 07102	35	40	60	57.1	22.8	12.5	58.5	17.6
55	P ₄	BG 3013	34	40	61.7	59.0	5.8	7.5	60.3	6.6
56	P ₅	PBG 5	45	44	80.0	68.1	20.0	13.6	74.0	16.8
57	P ₆	GNG 1958 (AVT-2)	44	42	61.3	66.6	15.9	9.5	63.9	12.7
58	P ₇	GL 27104	51	50	64.7	80.0	29.4	26.0	72.3	27.7
59	P ₈	GNG 1581	36	42	84.5	86.4	22.2	19.0	85.5	20.6
60	P ₉	GNG 1936	48	38	64.5	81.5	20.8	23.6	73.0	22.2
61	P ₁₀	GL 27091	44	42	63.6	60.4	15.9	23.8	62.0	19.8
62	P ₁₁	H 07-163	41	38	73.1	78.0	9.8	10.5	75.5	10.1
63	P ₁₂	GNG 469	25	28	64.0	62.5	12.0	14.2	63.2	13.1
64	P ₁₃	GL 26054	40	36	85.0	80.0	5.0	8.3	82.2	6.6
65	P ₁₄	GNG 1999	41	42	78.0	79.0	24.3	19.0	78.5	21.6
66	P ₅₁	BG 3032	42	37	88.0	77.1	33.3	35.1	82.5	34.2
67	P ₅₂	Phule G 0204-16	50	46	40.5	35.2	30.0	32.6	37.8	31.3
68	P ₅₃	GJG 0922	54	47	11.1	14.2	31.4	28.5	12.6	29.9
69	P ₅₄	BGD 1071	51	46	98.0	90.2	35.5	59.0	94.1	37.2
70	P ₅₅	CSJ 697	52	50	94.2	95.3	1.9	2.0	94.7	1.9
71	P ₅₆	H 08-25	40	43	57.6	62.7	4.6	11.6	60.1	8.1
72	P ₅₇	GNG 0921	42	62	11.9	9.6	11.9	8.0	10.7	9.9
73	P ₅₈	Phule G 06102	37	40	45.9	40.2	21.6	30.0	43.0	25.8
74	P ₅₉	IPC 06-127	38	50	23.8	20.2	15.7	12.5	22.0	14.1
75	P ₆₀	JG 28	35	40	35.2	37.5	18.5	20.0	36.3	19.2
76	P ₆₁	NDG 1105	33	40	84.8	97.5	9.0	7.5	91.1	8.2
77	P ₆₂	GNG 2066	49	47	97.7	98.2	8.5	10.5	98.0	9.5
78	P ₆₃	JG 27	52	53	71.1	73.2	26.9	18.8	72.1	22.8
79	P ₆₄	IPC 07-56	44	40	11.3	14.2	15.9	21.5	12.7	18.7
80	P ₆₅	PG 0101	48	39	95.8	89.2	2.0	2.5	92.5	2.2
81	P ₆₆	RVSSG 8	17	19	52.9	56.6	52.9	47.3	54.7	25.0
82	P ₆₇	IPC 2006-84	42	48	52.3	57.5	4.7	10.4	54.9	7.5
83	P ₆₈	JG 14	41	49	59.5	48.5	9.7	8.1	53.5	8.9
84	P ₆₉	BGD 1072	40	46	90.9	89.2	7.5	6.5	90.0	7.0
85	P ₇₀	IPC 2006-77	46	52	21.7	24.1	10.8	19.2	22.9	15.0
86	P ₇₁	RSG 963	44	48	90.5	94.2	15.2	18.2	92.3	16.7
87	P ₇₂	H 08-75	42	46	21.4	28.2	4.7	2.1	24.8	3.4
88	P ₇₃	CSJ 753	41	47	94.5	95.3	2.4	3.2	94.9	2.8

89	P ₇₄	BG 3031	45	49	86.6	95.5	8.8	8.1	91	8.4
90	P ₇₅	GNG 22081	34	38	54.2	50.2	2.8	2.6	52.1	2.7
91	P ₄₈	GNG 1991	33	43	90.9	83.7	16.2	18.6	87.3	17.4
92	P ₄₉	RSG 963	39	44	94.2	87.7	25.3	30.2	90.9	27.7
93	P ₅₀	GNG 1995	30	40	80.0	70.5	20.0	20.0	75.2	20.0
94	P ₈₁	RSG 888	50	48	94.5	97.0	25.0	20.2	95.7	22.6
95	P ₈₂	CSJ 515	42	44	38.0	35.3	3.3	4.5	36.6	3.9
96	P ₈₃	RSG 931	46	50	96.6	92.0	10.5	12.3	94.3	11.4
97	P ₈₄	GNG 1996	45	50	68.8	80.1	33.3	26.6	74.4	29.9
98	P ₈₅	BGM 572	40	45	94.5	95.0	30.2	26.5	94.7	28.3
99	JG315(R) check		48	45	5.6	4.8	8.2	6.5	7.5	6.5
100	BG212(S) check		47	48	95.0	90.0	100	100		
	LSI								79.1	19.2

R= Resistant S=Susceptible

5.2.3 Vascular wilt (kabuli genotype)

Out of the 61 (Kabuli) entries evaluated one HK 08-212, were found resistant and 4 entries i.e. CSJK 74, CSJK 72, JGK 16, CSJK 66, were found moderately resistant .The disease incidence ranged from 5.5 to 94.8 percent. The mortality was 11.5 per cent in resistant check JG 315 and 69.0 per cent susceptible check BG 212 .

5.2.4 Dry root rot (kabuli genotype)

The data presented in table 9 indicates that of the 61 Kabuli entries 18 i.e. CSJK 54, GNG 2104, HK 08-212, CSJK 74, GNG 2047, IPC 06-143, GLK 26162, HK 06-152, CSJK 66, GNG 1969, GLK 26155, BG 3012, HK 08-206, JGK 13, IPCK 08-120, CSJK 70, JGK 19, BG 3027 were resistant and 23 entries namely (JGK 2005-301, CSJK 6, JGK 2003-04, Phule G 09305, CSJK-1, GNG 2034, vihar (Phule G 95311), CSJK 68, RVSSG 11. GNG 2112, Kripa (Phule G 0517), HK 08-206, JGK 18, JGK 16, IPCK 08-136, Phule G 09316, HK 6-163, HK 07-234, Phule G 0027, CSJK 66, BG 3027) were moderately resistant. The disease incidence ranged from 4.1 to 69.4 percent.

Table 07 ; Screening of chickpea Kabuli entries against vascular wilt and dry root rot under sick field condition.

S.No.	Entry code	E.name	Total plants		Percent mortality (%)				Average (%)	
					Wilt		Dry root rot		Wilt	Dry root rot
			R I	R II	R I	R II	R I	R II	R I	R II
1	P ₇₁	GLK 28127	11	10	18.1	24.4	18.1	30.0	21.2	24.0
2	P ₇₂	Kripa (Phule G 0517)	6	9	66.6	62.2	66.6	60.2	64.4	63.4
3	P ₇₃	GLK 26167	17	14	85.4	78.4	40.5	45.3	81.9	42.9
4	P ₁₅₁	Vihar	9	7	60.4	57.1	11.1	14.2	58.7	12.6
5	P ₁₅₂	HK 08-	18	25	34.4	38.2	16.6	12.0	36.3	14.3
6	P ₁₅₃	CSJK 68	17	22	64.5	68.1	15.2	18.3	66.3	16.7
7	P ₁₅₄	BG 3025	20	23	40.3	34.7	20.0	25.5	37.5	22.7
8	P ₁₅₅	RVSSG 11	28	34	83.3	76.4	14.2	19.5	79.8	16.8
9	P ₁₅₆	GNG 2112	20	23	80.2	73.7	12.5	14.5	77.0	13.5
10	P ₁₅₇	HK 06-171	19	21	42.1	47.6	26.3	23.8	44.8	25.0
11	P ₁₅₈	IPCK 07-62	8	14	62.5	71.4	50.0	38.5	66.9	44.2
12	P ₁₅₉	RVSSG 12	7	9	42.0	40.8	57.1	52.5	41.4	54.8
13	P ₁₆₀	Phule G 09311	21	25	38.0	35.3	50.2	51.3	36.6	50.7
14	P ₁₆₁	CSJK 54	23	28	65.2	60.7	4.3	7.1	62.9	5.7
15	P ₁₆₂	Phule G 09316	25	30	96.2	97.2	12.5	18.5	96.7	15.5
16	P ₁₆₃	GNG 2104	28	32	56.2	45.6	3.5	6.3	50.9	4.8
17	P ₁₆₄	HK 08-212	26	28	3.8	7.2	3.8	7.1	5.5	5.4
18	P ₁₆₅	CSJK 74	29	33	16.6	19.2	8.5	9.7	17.9	9.1
19	P ₁₆₆	GNG 2047	30	34	95.0	91.1	3.3	11.7	93.0	7.5
20	P ₁₆₇	IPC 06-143	31	23	32.1	36.0	9.6	8.6	34.0	9.1
21	P ₁₆₈	HK 06-163	23	23	43.4	48.6	13.0	21.7	46.0	17.3
22	P ₁₃₀	SC-RS-1	7	6	94.3	95.3	28.5	33.3	94.8	30.9
23	P ₁₃₁	HK 94-134	9	7	39.0	40.3	33.3	29.3	39.6	31.3
24	P ₁₃₂	GNG 1888	3	4	93.0	89.2	31.2	40.3	91.1	51.5
25	P ₁₃₃	IPCK 113	7	5	91.2	93.2	28.5	20.0	92.2	24.2
26	P ₁₃₄	JGK 1	9	8	33.3	34.3	44.4	36.5	33.8	40.4
27	P ₁₃₅	HK 07-234	10	12	91.2	95.3	20.0	18.3	94.7	19.1
28	P ₁₃₆	Phule G	12	9	96.6	91.2	8.3	12.5	93.9	10.4
29	P ₁₃₇	CSJK 6	15	12	86.6	89.2	20.0	25.0	87.9	22.5
30	P ₁₃₈	IPC06-78	8	9	25.2	29.3	18.4	23.5	27.2	20.9
31	P ₁₃₉	CSJK 42R	9	7	86.2	78.2	44.4	48.5	82.2	46.4
32	P ₁₄₀	GNG 2034	14	17	78.6	72.3	14.2	11.7	75.4	12.9
33	P ₁₄₁	GLK 26162	16	18	68.7	86.2	6.2	5.5	77.4	5.8
34	P ₁₄₂	IPCK 06-56	13	16	30.3	34.2	23.0	18.7	32.4	20.8
35	P ₁₄₃	GNG 1969	9	7	80.2	78.2	22.2	28.5	79.2	25.3

36	P ₁₄₄	HK 06-152	17	10	64.7	66.6	5.8	8.2	65.6	7.0
37	P ₁₂₁	JGK 2005-301	7	8	62.3	64.0	14.2	20.3	63.1	17.2
38	P ₁₂₂	CSJK 66	18	20	70.0	73.8	5.5	11.5	71.9	8.5
39	P ₁₂₃	Phule G 0027	7	9	75.0	77.7	14.2	18.4	76.3	16.3
40	P ₁₂₄	JGK 2003-04	13	19	81.2	84.2	15.3	13.2	82.7	14.2
41	P ₁₂₅	CSJK-1	10	15	85.4	75.3	10.0	13.3	80.3	11.6
42	P ₁₂₆	GNG 1969	10	14	90.0	92.8	11.7	7.1	91.4	9.4
43	P ₁₂₇	GLK 26155	17	21	68.8	66.6	5.8	9.5	67.7	7.6
44	P ₁₇₅	Phule G 09305	16	15	43.7	50.0	13.3	15.0	46.8	14.1
45	P ₁₇₆	JGK 17	13	10	23.0	23.0	15.3	18.0	24.0	16.6
46	P ₁₇₇	IPCK 08-136	19	16	89.5	85.0	15.7	12.5	87.2	14.1
47	P ₁₇₈	BG 3012	30	28	80.0	82.0	6.6	3.5	81.0	5.0
48	P ₁₇₉	HK 08-206	22	18	31.8	37.1	4.5	7.4	34.4	5.9
49	P ₁₈₀	JGK 13	29	25	85.0	88.0	6.8	8.5	86.5	7.6
50	P ₁₈₁	Vihar (Phule G 95311)	16	14	37.5	65.6	12.5	14.2	51.5	13.3
51	P ₁₈₂	IPCK 08-120	8	11	40.0	45.0	10.5	9.0	42.5	9.7
52	P ₁₈₃	JGK 18	15	18	36.2	42.8	13.3	16.6	39.5	14.9
53	P ₁₈₄	AKG2002-1K	14	14	33.9	38.5	28.5	21.4	36.2	24.9
54	P ₁₈₅	CSJK 72	10	10	10.0	16.5	30.0	20.0	13.2	25.0
55	P ₁₈₆	JGK 16	12	10	16.6	20.0	16.6	20.0	18.3	18.3
56	P ₁₈₇	IPCK 08-130	9	8	22.5	26.2	22.2	18.5	48.7	20.3
57	P ₁₈₈	CSJK 70	25	28	64.5	69.2	8.0	10.7	66.8	9.3
58	P ₁₈₉	Kripa (Phule G0517)	22	18	18.1	26.2	9.0	16.6	22.1	12.8
59	P ₁₉₀	CSJK 66	7	12	14.2	20.5	14.2	16.6	17.3	15.4
60	P ₁₉₁	JGK 19	22	26	89.0	84.5	4.5	3.8	86.7	4.1
61	P ₁₉₂	BG 3027	25	24	30.5	32.0	8.0	6.5	31.2	7.2
62		JG 315 (R) check	35	33	10.5	12.5	15.0	18.5	11.5	16.2
63		BG 212(S) check	30	25	65.7	72.3	69.0	31.2	69.0	50.2
					L.S.I				46.3	13.2

R =Resistant, S = Susceptible

5.3.1 Dry root rot (desi genotype *in vitro*)

In order to find out the resistant genotype against dry root rot 98 chickpea (desi) entries were evaluated. In-vitro condition among them five entries viz GJG 0910, IPCK 06-78, CSJK-42, IPCK 06-56, GNG 1969 were found to be resistant. Exhibiting, less than 10 percent disease incidence whereas 42 entries namely (GL 27091, H 07-163, GJG 0907, RVSSG 9, CSJ 313, RVSSG 10, GNG 2065, Phule G 0204, GNG 0906, CSJ 0564, GNG 2064, AKG 1001, H 08-131, H 08-18, JG 23, BG 032, Phule G 0204-16, BGD

107, H 08-25, GNG 0921, Phule G 06 102, IPC 06-127, JG 28, NDG 1105, GNG 2066, IPC 06-127, JG 28, NDG 1105, GNG 2066, IPC 2006-77 ,RSG 963, CSJ 753, BG 3031, GNG 2081, RSG 888, CSJ 515, RSG 931, BGD 1070, Vijay , BGM 572, BCP 60, JG 25, GJG 0814, CSJ 513, H 08-93, GNG 2068) were found to be moderately resistant .The disease incidence ranged from 10.1 to 20 percent.(Table 8)

Table 08. Evaluation of chickpea desi entries against dry root rot by using blotter paper technique

S. NO.	EN. Code	En. Name	Rating	Percentage mortality	Reaction
1	P ₁	NBeG 49	8-9	>40.1	H.S.
2	P ₂	GJG 0809	8-9	>40.1	H.S.
3	P ₃	Phule G 07102	8-9	>40.1	H.S.
4	P ₄	BG 3013	8-9	>40.1	H.S.
5	P ₅	PBG 5	8-9	>40.1	H.S.
6	P ₆	GNG 1958 (AVT-2)	8-9	>40.1	H.S.
7	P ₇	GL 27104	8-9	>40.1	H.S.
8	P ₈	GNG 1581	4-5	20.1-30.0	H.S.
9	P ₉	GNG 1936	4-5	20.1-30.0	T
10	P ₁₀	GL 27091	2-3	10.1-20.0	M.R.
11	P ₁₁	H 07-163	2-3	10.1-20.0	M.R.
12	P ₁₂	GNG 469	8-9	>40.1	H.S.
13	P ₁₃	GL 26054	8-9	>40.1	H.S.
14	P ₁₄	GNG 1999	6-7	30.1-40.0	H.S.
15	P ₂₁	BG 3024	4-5	20.1-30.0	S
16	P ₂₂	BDNG 2010-1	6-7	30.1-40.0	T
17	P ₂₃	BGD 1068	6-7	30.1-40.0	S
18.	P ₂₄	JG 24	6-7	30.1-40.0	S

19.	P ₂₅	PG 0105	6-7	30.1-40.0	S
20.	P ₂₆	Phule G 0215-2	6-7	30.1-40.0	S
21.	P ₂₇	RVSSG 9	6-7	30.1-40.0	S
22.	P ₂₈	GJG 0907	2-3	10.1-20.0	MR
23.	P ₂₉	CSJ 313	2-3	10.1-20.0	MR
24.	P ₃₀	RVSSG 10	2-3	10.1-20.0	MR
25.	P ₃₁	GNG 2065	2-3	10.1-20.0	MR
26.	P ₃₂	Phule G 0204-4	2-3	10.1-20.0	MR
27.	P ₃₃	GJG 0906	2-3	10.1-20.0	MR
28.	P ₃₄	CSJ 0564	2-3	10.1-20.0	MR
29.	P ₃₅	GNG 2064	2-3	10.1-20.0	MR
30.	P ₃₆	GJG 0809	1	0.0-10.0	R
31.	P ₃₇	BG 3023	2-3	10.0-20.0	MR
32.	P ₃₈	CSJ 303	1	0.0-10.0	R
33.	P ₃₉	AKG 1001	2-3	10.1-20.0	MR
34.	P ₄₀	IPC 07-19	4-5	20.1-30.0	T
35.	P ₄₁	H 08-13	2-3	10.1-20.0	MR
36.	P ₄₂	BG 3004	1	0.0-10.0	R
37.	P ₄₃	IPC 08-11	4-5	20.1-30.0	T
38.	P ₄₄	H 08-18	2-3	10.1-20.0	MR
39.	P ₄₅	JG 23	2-3	10.1-20.0	MR
40.	P ₄₈	GNG 1991	4-5	20.1-30.0	T
41.	P ₄₉	RSG 963	4-5	20.1-30.0	T
42.	P ₅₀	GNG 1995	8-9	>40.1	H.S
43.	P ₅₁	BG 3032	2-3	10.1-20.0	MR

44.	P ₅₂	Phule G 0204-16	2-3	10.1-20.0	MR
45	P ₅₃	GJG 0922	4-5	20.1-30.0	T
46	P ₅₄	BGD 1071	2-3	10.1-30.0	MR
47	P ₅₅	CSJ 697	4-5	20.1-30.0	T
48	P ₅₆	H 08-25	2-3	10.1-20.0	MR
49	P ₅₇	GNG 0921	2-3	10.1-20.0	MR
50	P ₅₈	Phule G 06102	2-3	10.1-20.0	MR
51	P ₅₉	IPC 06-127	2-3	10.1-20.0	MR
52	P ₆₀	JG 28	2-3	10.1-20.0	MR
53	P ₆₁	NDG 1105	2-3	10.1-20.0	MR
54	P ₆₂	GNG 2066	2-3	10.1-20.0	MR
55	P ₆₃	JG 27	4-5	20.1-30.0	T
56	P ₆₄	IPC 07-56	4-5	20.1-30.0	T
57	P ₆₅	PG 0101	4-5	20.1-30.0	T
58	P ₆₆	RVSSG 8	4-5	20.1-30.0	T
59	P ₆₇	IPC 2006-84	4-5	20.1-30.0	T
60	P ₆₈	JG 14	4-5	20.1-30	T
61	P ₆₉	BGD 1072	4-5	20.1-30.0	T
62	P ₇₀	IPC 2006-77	2-3	10.1-20.0	MR
63	P ₇₁	RSG 963	2-3	10.1-20.0	MR
64	P ₇₂	H 08-75	1	0.0-10.0	R
65	P ₇₃	CSJ 753	2-3	10.1-20.0	MR
66	P ₇₄	BG 3031	2-3	10.1-20.0	MR
67	P ₇₅	GNG 2081	2-3	10.1-20.0	MR

68	P ₈₁	RSG 888	2-3	10.1-20.0	MR
69	P ₈₂	CSJ 515	2-3	10.1-20.0	MR
70	P ₈₃	RSG 931	2-3	10.1-20.0	MR
71	P ₈₄	GNG 1996	4-5	20.1-30.0	T
72	P ₈₅	BGM 572	6-7	30.1-40.0	S
73	P ₉₁	BGD 1070	2-3	10.1-20.0	MR
74	P ₉₂	GJG 0910	1	0.0-10.0	R
75	P ₉₃	Vijay	2-3	10.1-20.0	MR
76	P ₉₄	BGM 572	2-3	10.1-20.0	MR
77	P ₉₅	BCP 60	2-3	10.1-20.0	MR
78	P ₉₆	CSJ 730	4-5	20.1-30.0	T
79	P ₉₇	JG 25	2-3	10.1-20.0	MR
80	P ₉₈	GJG 0814	2-3	10.1-20.0	MR
81	P ₉₉	CSJ 513	2-3	10.1-20.0	MR
82	P ₁₀₀	JG 26	4-5	20.1-30.0	T
83	P ₁₀₁	H 08-93	2-3	10.1-20.0	MR
84	P ₁₀₂	BG 3030	4-5	20.1-30.0	T
85	P ₁₀₃	GJG 0904	4-5	20.1-30.0	T
86	P ₁₀₄	BGD 1069	4-5	20.1-30.	T
87	P ₁₀₅	RSG 931	4-5	20.1-30.0	T
88	P ₁₀₆	GNG 2002	4-5	20.1-30.0	T
89	P ₁₀₇	GJG 0825	6-7	30.1-40.0	S
90	P ₁₀₈	H 08-71	4-5	20.1-30.0	T
91	P ₁₀₉	IPC 08-68	4-5	20.1-30.0	T
92	P ₁₁₀	Phule G 09103	6-7	30.1-40.0	S
93	P ₁₁₁	GNG 2068	2-3	20.1-30.0	MR

94	P ₁₁₂	RSG 888	4-5	20.1-30.0	T
95	P ₁₁₃	BG 3029	4-5	20.1-30.0	T
96	P ₁₁₄	GNG 2085	4-5	20.1-30.0	T
97	P ₁₁₅	IPC 07-09	4-5	20.1-30.0	T
98	P ₁₁₆	BGD 1061	4-5	20.1-30.0	T
99	JG 315 (R) Check		1-2	0.0 -10.0	R
100	BG212 (S) Check		8-9	>40	HS

R=Resistant, MR= Moderately resistant= Tolerant, S=Susceptible, HR= Highly susceptible

5.3.2 Dry root rot

In this trial 61 (Kabuli) entries were evaluated, out of which 14 lines viz GNG 1969, CSJK 68, Vihar (PhuleG95311), RVSSG 11, GNG 2112, HK 06-171, Kripa (Phule G 0517), IPCK 08-136, BG 3012 ,HK 08-206, JGK 13, AKG 2002-1K, IPCK 08-130, BG 3027 were found resistant whereas 25 entries IPCK 06-78, CSJK 42, IPCK 0656,GNG 1969 152, HK 08231, BG 3025, GLK 28127, GLK 26167, IPCK 08120,CSJK 72, CSJK 66, GJK 19,JGK 2005-301, vihar, CSJK-6, JGK-2003-304, CSJK-1, SC-RS-1, HK 94134, GNG 1888, JGK 1,HK 07-234, CSJK 6,CSJK 42, GNG 2034.were found moderately resistant showing 10.1-20 percent disease incidence. (Table 9)

Table: 09 Evaluation of chickpea entries against dry root rot by using blotter paper technique.

Kabuli En.61

S. NO.	En Code	En.name	Rating	Dry root rot Percentage	Reaction
1.	P ₁₂₁	JGK 2005-301	2-3	10.1-20	MR
2.	P ₁₂₂	CSJK 6	2-3	10.1-20	MR
3.	P ₁₂₃	Phule G 0027	4-5	20.1-30	T
4.	P ₁₂₄	JGK 2003-304	2-3	10.1-20	MR
5.	P ₁₂₅	CSJK-1	2-3	10.1-20	MR
6.	P ₁₂₆	GNG 1969	4-5	20.1-30	T
7.	P ₁₂₇	GLK 26155	4-5	20.1-30	T
8.	P ₁₃₀	SC-RS-1	2-3	10.1-20	MR
9.	P ₁₃₁	HK 94-134	2-3	10.1-20	MR
10.	P ₁₃₂	GNG 1888	2-3	10.1-20	MR
11.	P ₁₃₃	IPCK 113	4-5	20.1-30	T
12.	P ₁₃₄	JGK 1	2-3	10.1-20	MR
13.	P ₁₃₅	HK 07-234	2-3	10.1-20	MR
14.	P ₁₃₆	Phule G 04305	4-5	20.1-30	T
15.	P ₁₃₇	CSJK 6	4-5	10.1-20	MR
16.	P ₁₃₈	SC-RS-1	1	0.0-10	R
17.	P ₁₃₉	HK 94-134	1	0.0-10	R
18.	P ₁₄₀	GNG 1888	2-3	10.1-20	MR
19.	P ₁₄₁	IPCK 113	2-3	10.1-20	MR
20.	P ₁₄₂	JGK 1	1	0.0-10	R
20.	P ₁₄₂	JGK 1	1	0.0-10	R
21.	P ₁₄₃	HK 07-234	2-3	10.1-20	MR
22.	P ₁₄₄	Phule G 04305	4-5	20.1-30	T

23	P ₁₅₁	Vihar	2-3	10.1-20	MR
24	P ₁₅₂	HK 08-231	1	0.0-10	R
25	P ₁₅₃	CSJK 68	2-3	10.1-20	MR
26	P ₁₅₄	BG 3025	1	0.0-10	R
27	P ₁₅₅	RVSSG 11	2-3	10.1-20	MR
28	P ₁₅₆	GNG 2112	2-3	10.1-20	MR
29	P ₁₅₇	HK 06-171	2-3	10.1-20	MR
30	P ₁₅₈	IPCK 07-62	6-7	30.1-40	S
31	P ₁₅₉	RVSSG 12	4-5	20.1-30	T
32	P ₁₆₀	Phule G 09311	4-5	20.1-30	T
33	P ₁₆₁	CSJK 54	4-5	20.1-30	T
34	P ₁₆₂	Phule G 09316	4-5	20.1-30	T
35	P ₁₆₃	GNG 2104	4-5	20.1-30	T
36	P ₁₆₄	HK 08-212	4-5	20.1-30	T
37	P ₁₆₅	CSJK 74	6-7	30.1-40	S
38	P ₁₆₆	GNG 2047	6-7	30.1-40	S
39	P ₁₆₇	IPC 06-143	8-9	>40.1	HR
40	P ₁₆₈	HK 06-163	6-7	30.1-40	S
41	P ₁₇₁	Phule G 09311	1	0.0-10	R
42	P ₁₇₂	CSJK 54	2	10.1-20	MR
43	P ₁₇₃	Phule G 09316	1	0.0-10	R
44	P ₁₇₅	Phule G 09305	4-5	20.1-30	T
45	P ₁₇₆	JGK 17	4-5	20.1-30	T
46	P ₁₇₇	IPCK 08-136	2-3	10.1-20	MR
47	P ₁₇₈	BG 3012	2-3	10.1-20	MR
48	P ₁₇₉	HK 08-206	2-3	10.1-20	MR

49	P ₁₈₀	JGK 13	2-3	10.1-20	MR
50	P ₁₈₁	Vihar (Phule G 95311)	1	0.0-10	R
51	P ₁₈₂	IPCK 08-120	1	0.0-10	R
52	P ₁₈₃	JGK 18	1	0.0-10	R
53	P ₁₈₄	AKG 2002-1K	4-5	20.1-30	T
54	P ₁₈₅	CSJK 72	1	0.0-10	R
55	P ₁₈₆	JGK 16	2	10.1-20	MR
56	P ₁₈₇	IPCK 08-130	3	10.1-20	MR
57	P ₁₈₈	CSJK 70	4-5	20.1-30	T
58	P ₁₈₉	Kripa (Phule G 0517)	4-5	20.1-30	T
59	P ₁₉₀	CSJK 66	1	0.0-10	R
60	P ₁₉₁	JGK 19	1	0.0-10	R
61	P ₁₉₂	BG 3027	2-3	10.1-20	MR
64	JG 315 (R) check		1	0.0-10	R
63	JG 62 (S) check		8-9	>40	HS

R=Resistant, MR= moderately resistant= Tolerant, S=Susceptible, HR= highly susceptible

Table 10 Comparative performance of chickpea entries against collar rot, vascular wilt and dry root rot.

Mortality Percent	Collar rot	Vascular wilt	Dry root rot.	Grading
0-10%	Desi—GJG0809, GL27104, GNG 1581, GL2709, IPC 06127, NDG1105, PG0105, GNG2065, Phule G0204, BG 3023, IPC0719, BG 3004, GNG 2068, IPC 07-09. Kabuli – IPCK 08-120, CSJK 68, RVSSG 11, GNG2112, HK 06-171, CSJK 54, Phule G 09316, HK 08-212, CSJK 74, IPC 06-143, Phule G 04305, GLK 26162, HK 06-152.	Desi – BDNG 2010-1, GNG 2064, IPC 08-11, H 08-93, BG 3030, BGD 1069, GNG 2002, GNG2085, GNG0921 Kabuli HK-08-212, GLK 28-127.	Desi –GJG 0809, CSJ 303, BG 3004, GJG 0910, IPCK 06-56. Kabuli— IPCK 06-78, CSJK 4, IPCK 06 -56, HK 08-231, BG 3025, GLK 28127, GLK 26167, Vihar (Phule G 95311), IPCK 08-120, JGK 19, CSJK 66, JGK18, CSJK72.	Resistance
10-20%	Desi— Phule G 07102, H 07163, GL2605, GNG1999, JG24, Phule G02152, GJG0907, CSJ313, RVSS G10, GJG0809, GJG 0814, H 08-71, CSJ515, GNG1991, CSJ697, GNG 0921. Kabuli -HK 08 -231, CSJK 74, Phule G 04305, GNG 1969, IPCK 06-56, CSJK 6.	Desi - JG 24, AKG 1001, BG 3004, JG 25, CSJ 513, H 08-71, BGD1061, GJG 092, IPC07-56. Kabuli -- CSJK 74, JGK 16, CSJG 72, BG 3027.	Desi- GL 27091, H 07-163, GJG 0907, RVSSG 9, CSJ 313, RVSSG 10, GNG 2065, Phule G 0204-4, GIG 0906, CSJ 0564, GNG 2064, AKG 1001, H 08-13, H 08-18, JG 23, BG 3032, Phule G 0204-16, BGD 1071, H 08-25, GNG 0921, Phule G06102, IPC 06-127, JG 28, NDG 1105, GNG 2066, IPC 2006-77, RSG 963, CSJ 753, BG 3031, GNG2081, RSG888, CSJ515, RSG931, BGD1070, Vijay, BGM 572, BCP60, JG25, GJG0814, CSJ 513, H 08-93, GNG 2068. Kabuli - JGK 2005-301, CSJK 6, JGK 2003-304, CSJK1, SC-RS 1, HK 94-134, HK 07-234, JGK 1, CSJK 6, CSJK 42, GNG 2034, GNG 1969, Vihar, CSJK 68, RVSSG 11, GNG 2112, HK06-171, kripa (Phule G 0517), IPCK 08-136, JGK 13, HK 08-206, JGK, JGK13, JGK16, IPCK08-130, BG 3027.	Moderately resistance

20-30%	Desi -BG 3031, JG -28, JG -27,PG 0101, BG3024,BDNG 2010, RVSSG 9, CSJ 303, BG 3004 CSJ 730, RSG 931. Kabuli-IPCK07-62,RVSSG 12, GNG 2047, GLK 28127 .	Desi - GJG 0814, IPC 08 68, PHULE 9103, IPC 07-09,Phule G 0204-4,IPC 6-127, IPC 2006-77, HK 08-75 Kabuli- GLK 28127, IPCK 06-78,JGK-17,Kripa(Phule G 0517)	Desi- GNG1581,GNG1936,BG3024, IPC07-19,IPC08-11,GNG1991,RSG 963,GJG0922,CSJ697,JG27,IPC 07-56,PG0101,RVSSG8,IPC 2006-84,JG4, BGD 1072,GNG 1996, CSJ 730, BG3030, GJG0904, JG26,BGD1069, RSG 931, GNG 2002,H08-71, Phule G 0903, RSG 888, BG 3029, GNG 2085, IPC 07-09, BGD 1061. Kabuli-PhuleG 0027,GNG1969,GLK 26155, IPCK113RY ,PhuleG04305,HK06152RVSSG12,PhuleG09311, CSJK5,PhuleG-09316,GNG2104,HR08-212,phule G09305, JGK17, AKG 2002-1K, CSJK70, Kripa(Phule G 0517)	Tolerant
30-40%	Desi- PBG5,GNG469,GJG0922,BGD1071, IPC200684,IPC200677,CSJ753,BG3 031,BGD1068,CSJ0564,GNG2064, AKG1001, IPC-08-11,RSG931,Phule 09103,RSG888 . Kabuli-HR-08-206,JGK-18, AKG2002-1K,SC– RS -1	Desi – GJG 0825, Phule G0215-2, JG -28, BG 3032, CSJ515. Kabul- HK94134,JGK1,IPCK0656,HK0 8-206,JGK18,AKG2002-1K,BG3027,HK08231,BG3025, PhuleG09311,IPC06-143.	Desi -GNG 1999,BDNG 2010-1 BDG 1068, JG 24,PG 0105,Phule 0215-2, RVSSG 9,GNG 1995,BGM 572, Phule G09103 Kabuli-CSJK 74,GNG 2047,HK 06-163.	Susceptible

40% above	Desi- BG3013,GNG1936,H0825,IPC0756,JG14,H0875,GNG2081,PhuleG06102,GJG0906,JG23,BGD1070,GJG0910,Vijay,BGM572,BCP60,JG25,CSJ513,JG26,H-08-93,BG3030,GJG0904,BGD1069,GNG2002,GJG0825,IPC0868,BG3029,GNG2085,BGD1061,RSG888, GNG1996,BGM572. Kabuli PhuleG-09305,JGK17,BG3012,HK08-206,JGK-13,Vihar (Phule G95311) ,CSJK72,JGK16,IPCK08130,CSJK70,Kripa(PhuleG0577),CSJK66,JGK9,BGB3027,Vihar,JGK2005301,CSJK66,PhuleG0027,JGK200304,CSJK1,GNG1969,GLK26155,HK94-134,GNG1888,IPCK113,JGK1,HK07234,CSJK42,GNG2034,kripa (phule G0517),GLK 26167.	Desi- BGD1070,Vijay,BGM572,CSJ730,JG26,GJG0904,GNG2068,RS G888,BG3029,BGB3024,BGD1068,PG0105,RVSSG9,GJG0907,CSJ313,RVSSG10,GNG2065,GJG0906,CSJ0564,GJG0809,BG3023,CSJ03,H0818,JG23,NBeG9,GJG0809,PhuleG07102,BG3013,PBG5,GNG158,GL27091,GNG1581,GNG193GL27091,H07163,GNG469,GL26054,GNG1999,BG3032,BGD1071,CSJ697,H0825,PhuleG06102,NDG1105,GNG2066,JG27,PG0101,RVSSG8,IPC2006-84,JG74, RSG 963,CSJ 753 BG3031,GNG2081,RSG888,RSG931,GNG1996,BGM572,GNG1991,RSG 963,GJG1995. Kabuli kripa (Phule G 0517) GLK26167,Vihar,CSJK68,VSSG11,GNG2112,HK06171,IPCK0762,RVSSG12,PhuleG09311,CSJK54,RGNG1888,IPCK113,HK07-234,Phule-G-04305,CSJK6,CSJK42,GNG2034,GLK2616,GNG1969,HK06152,JGK2005-301,CSJK1,GNG1969,GLK26155,CSJK6,PhuleG0027,JGK2003.304,phuleG09305,IPCK08136,BG3012,JGK13,Vihar(PhuleG95311),IPCK0820,IPCK08130,CSJK70,JGK19.	Desi - NBEG 49,GJG0809, Phule G 07102 , BG 3013, PBG 5,GNG1958,GL27104,GNG 469,GL26054. Kabuli -IPC06-143.	Highly susceptible
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5.3.2 Performance of chickpea entries exhibiting dual and multiple disease resistant against collar rot, vascular wilt and dry root rot.

In the present investigation one genotype 159 desi and kabuli genotype were screened against collar rot vascular wilt and dry root rot in multiple disease sick plot at Jabalpur. Each genotype was alternated with a single row of susceptible (JG 62) and resistant control (JG 315). Genotype namely GLK 28127 exhibited resistant against all the three pathogens. Genotype BGD 1071 (desi) and Phule G 09316(Kabuli) showed resistant reaction, whereas JG 24 ,H 0871 exhibited moderately resistant reaction against collar rot and vascular wilt. genotype IPCK 0678 (Kabuli exhibited resistant reaction whereas) Genotype H07-163, GJG 0907,CSJ (DESI) and CSJK-6 (Kabuli exhibited moderately resistant reaction dry root and possessing resistance to a single disease or for more than one disease may be utilized in breeding programme for developing high yielding cultivars with multiple disease resistance.

Table 11; Performance of chickpea entries exhibiting dual and multiple Disease resistant against collar rot, vascular wilt and dry root rot.

(i) Collar rot and Vascular wilt

Grading	Entries
Resistant (R)	Desi -BGD1071 Kabuli – Phule G 09316
Moderately resistant (MR)	Desi -JG24,H08-71 Kabuli - NIL

(ii) Dry root rot and collar rot

Grading	Entries
Resistant (R)	Desi -NIL Kabuli -IPCK 06-78
Moderately resistant (MR)	Desi H 07-163, JG 24, GJG 0907, CSJ 313. Kabuli –CSJK 6

(iii) - Vascular wilt and dry root rot.

Grading	Entries
Resistant (R)	Desi -BDNG 2010-1,GNG 2064,BG 3004 RIPC 08-11, H 08-93, BGD 1069,GNG2002 GNG0921. Kabuli -Phule G 04305, GLK 26162 , HK 06-152, HK 08-212
Moderately resistant (MR)	Desi - IPC 08-11, BCP 60, CSJ 513, BG 3030, H 08-71 Kabuli –CSJK 74, JGK 16.

DISCUSSION

Chickpea is an important protein rich crop of the world. It suffers heavily due to biotic and abiotic stresses. Among biotic, soil borne pathogens are most important. Vascular wilt, dry root rot and collar rot are the most devastating diseases in India. They occur from seedling to pod formation stage and are seed as well as soil borne in nature. Losses due to vascular wilt have been reported up to 60 per cent from Madhya Pradesh (Gupta *et al.*, 1987). Maurya *et al.* (2008) reported that collar rot (*Sclerotium rolfsii*) of chickpea is one of the serious soil borne diseases of fungal origin due to which 10-30 percent yield loss is recorded annually.

The incidence of root rot ranging from 3.5 to 20.6 per cent from 30 villages of northern Madhya Pradesh was reported by Gupta *et al.* (1983). Studies on symptoms of wilting at different stages of crop growth in disease sick plot revealed that the affected plants showed typical wilt symptoms like drooping of petioles and rachis alongwith leaflets and on longitudinal splitting of root showed brown to black discoloration in internal tissues. Massahebi (1968) also reported that the wilt symptoms as general yellowing of leaves and discoloration of vascular elements. Collar rot disease was found to occur mainly at seedling stage of crop growth (15-45 days). The affected plants turn yellow and die. The seedlings found to collapsed and show rotting at the collar region and below.

According to Mehrotra and Aneja (1990) first symptom of dry root rot disease appears in form of yellowing of leaves, within a day or two such leaves drop and in the course of next two or three days leading to death of plant. The leaves and stem of the affected plant are usually straw colored. The tap root is dark and quite brittle in dry soil.

Isolations from chickpea plants infected of collar rot, vascular wilt and dry root rot made on potato dextrose agar medium showed association of *Sclerotium rolfsii*, *Fusarium oxysporum* f.sp. *ciceri* and *Rhizoctonia bataticola*.

Pathogenicity test for all the three pathogens conducted by using susceptible JG 62 (wilt and collar rot), BG 212 (dry root rot) and resistant check JG 315 and JG 63 through pot inoculation and blotter paper technique revealed the pathogenic nature Grewal *et al.*(1974) and Zote *et al.* (1990) and Dermiro *et al.*(1994) also reported pathogenic behavior of organism.

The progressive development of collar rot, vascular wilt and dry root rot were investigated in relation in environmental factors using a extra large seeded kabuli chickpea genotype (Kripa Phaule G 0517) and small seeded desi chickpea (CSJ 515) grown in multiple disease sick field alongwith susceptible checks JG 62 and BG 212 (collar rot, wilt and dry root rot). Maximum collar rot (28.2%) and vascular wilt (22.5%) in desi variety (CSJ 515) and in kabuli variety Kripa (Phule G 0517) has shown 27.4% collar rot and 25.2% vascular wilt was in the 52nd standard week i.e. 5th week of Dec. 2011. During that period the temperature ranged from 7.4 to 24.9^oC and relative humidity 24.0 to 89 percent. The maximum percent mortality due to dry root rot was 25.3 per cent in desi variety (CSJ 515) and 29.2 per cent in Kabuli variety Kripa (Phule G- 0577) noticed the 10th standered week i.e. 2nd week of March 2012. At this period temperature was ranged from 11.6 to 30.8^oC coupled with relative humidity 18 to 76 percent.

Gangwar, *et al.* (2009) recorded maximum wilt incidence (11.3-18.8%) when the average soil temperature ranged from 25-29.4^oC and soil moisture from 28.4 to 56.5 per cent findings are in confirmation to result obtained during the course of investigation .

High incidence of dry root rot in Southern Haryana has been reported from late October to Mid November than again increased in the month of February and March (Tripathi and Sharma 1983)

To find out the host resistance sources 98 chickpea (desi) entries and 61 kabuli entries were evaluated against dry root rot. Among them five (5 desi) entries viz GJG 0910, IPCK 06-78, CSJK-42, IPCK 06-56, GNG 1969 and 13 kabuli entries IPCK 06-78,CSJK 4, IPCK 06 -56, HK 08-231,BG 3025, GLK 28127, GLK 26167, Vihar (Phule G 95311), IPCK 08-120, JGK 19, CSJK 66, JGK18, CSJK 72 were found to be resistant.

Prajapati *et al.* (2003) also evaluated 45 chickpea cultivars for their resistance and recorded that to dry root rot among these, H 355, JG 315, Pusa 212, G 543, ICCC 32, BG 244, Phule G 5, ICC 2644, 10384, 11224, 10630 and 11332, ICCV 10, ICC 12441 and 12263, 81010 and 81002, Avrodhi, ILC 702 and ILC 200 cultivars were resistant.

Screening studies made by Gupta *et al.* (2012) conducted that out of 170 accessions, 68 genotypes exhibited resistant reaction (<10% mortality), out of which 26 are the promising lines namely (JG1-14, 2-125, 2-4-110, 14-11, 14-10, 2001-13, 2001-13, 2001-18, 2001-80, 2001-115, 2002-20, 2003-95, 2003-14-16, 2004-110, 210, 9605, 1-9, 99-115, 2001-04, 2003-14-2, JG 2000-07, JSC 37, MPJG89-11551, MPJG 89-9023, CSJ 592.

Pande *et al.* (2004) reported that among the 29 chickpea germplasm accessions, 10 cultivars and 8 advanced breeding lines for resistance to dry root rot, caused by *Rhizoctonia bataticola* under *in vitro* conditions. They observed one germplasm accession (ICC 14395), a cultivar (ICCV 2) and advanced breeding line were found resistant to dry root rot.

Similarly the entries evaluated against collar rot resulted that 15 of desi type (GJG 0809, GNG 1958, GL 27104, IPS 06127, PG 0105, GNG 2065, Phule G 0204, IPC 07-19, H-08-13, GNG 2068, IPC 07-09, GJG 0907, GL 27091, IPC 06-127, GNG 1581) and 13 in kabuli type which 15 entries namely CSJK 68, HK 06-171, phuleG-09311, IPCK 08 120, IPCK 06-78, CSJK-54, GLK 26162, H K 06 152, Phule G 09316, HK 08-212, CSJK 74, IPS-06-163, RVSSG 11, GNG 2112, IPS-06-143 were found resistant) were found resistant.

Possible sources of resistance to collar rot has been reported by other workers from different places on the basis of pot inoculation and field screening techniques (Karat *et al.* 1985, Chittle *et al.* 1990, Sugha *et al.* 1991, and Husain *et al.*, 2005)

Gupta and Babbar (2006) have reported resistant lines for wilt and dry root rot in desi and kabuli chickpea. Studies on vascular wilt resistant exhibited nine entries desi and two entries kabuli entries were found resistant. Chaudhary *et al.* (2007) screened 196 chickpea germplasm lines/cultivars for resistance to wilt disease in a wilt sick plot. None of the test line was found immune or highly resistant.

Dua *et al* (2008) have also reported identified promising chickpea lines had combined resistance against collar rot ,wilt and dry root rot.

SUMMARY, CONCLUSIONS AND SUGGESTIONS FOR FURTHER WORK

SUMMARY

Chickpea is the one of the third most important grain legume protein rich crop in world. It is affected by several seed and soil borne diseases in Madhya Pradesh. Major losses in chickpea yield are attributed due to soil borne pathogens.

Among them collar rot, vascular wilt and dry root rot are important diseases and has been observed at seedling, flowering and podding stage of the crop. Diagnosis of the disease symptoms at different stage of crop growth revealed that the disease was found to be vascular wilt caused by *Fusarium oxysporum* f.sp. *ciceri*, collar rot by *Sclerotium rolfsii* and dry root rot due to *Rhizoctonia bataticola*.

The pathogen was indentified on the basis of morphological characters on PDA. The pathogenicity test performed by using pot inoculation and blotter paper techniques proved their pathogenic nature. The progressive development of collar rot, vascular wilt and dry root rot under field and net house condition were investigated in relation in environmental factors by using a extra large seeded kabuli chickpea genotype Kripa(Phule G 0517) and small seeded desi chickpea (CSJ 515) grown in multiple disease sick field alongwith susceptible checks JG 62 and BG 212 (collar rot, wilt and dry root rot) led to conclusion that maximum collar rot (28.2%) and vascular wilt (22.5%) incidence has been observed in the 52th standard week i.e. 5th week of Dec. 2011. During that period the temperature ranged from 7.4⁰c to 24.9⁰c and relative humidity, 24.0 to 89 percent which simultaneously lower down in both desi as well as kabuli chickpea.

On contrary to this, highest percent mortality due to dry root rot was observed in the 10th standard week i.e. 2nd week of March 2012 when the temperature was ranging from 11.6-30.8⁰C coupled with 18 to 76 percent

relative humidity. However, the disease initiated when the temperature was $>27^{\circ}\text{C}$ with the decrease in relative humidity too.

Studies on host resistance against collar rot, vascular wilt and dry root rot under wilt sick field and different inoculation techniques employed revealed that among 159 entries (98 desi and kabuli 61) screened for locating the resistant sources 9 entries. viz. BDNG 2010-I, GNG-2064, IPC-08-11, H08-93, BG-3030, BG-D 1069, GNG-2002, GNG-2085, GNG-0921 of desi type and 2 kabuli type (HK-08-212, GLK 28-127) were found resistant to wilt.

Fourteen entries of desi chickpea viz GJG 0809, GNG 1958, GL 27104, IPS 06127, PG 0105, GNG 2065, Phule G 0204, IPC 07-19, H-08-13, GNG 2068, IPC 07-09, GJG 0907, GL 27091, IPC 06-127, GNG 1581 and 15 Kabuli type entries namely CSJK 68, HK 06-171, phuleG-09311, IPCK 08 120, IPCK 06-78, CSJK-54, GLK 26162, H K 06 152, Phul e G 09316, HK 08-212, CSJK 74, IPS-06-163, RVSSG 11, GNG 2112, IPS-06-143 were found resistant) were found resistant to collar rot.

Four entries viz. GJG 09 809, CSJ 303, GJG-0910, IPCK 06-56 of desi type and 13 Kabuli type (IPCK 06-78, CSJK 4, IPCK06-56 HK08-231, BG3025, GLK 28127, GLK 26167, Vihar (phule G 95311) IPCK 08-120, JGK 19, CSJK 66, JGK 18, CSJK 72, CSJK 7) were found resistant to dry root rot.

Besides resistance for individual disease genotype viz BDNG 2010-1, GNG 2064, BG 3004, RIPC 08-11, H 08-93, BGD 1069, GNG 2002, GNG 0921. (vascular wilt and dry root rot.) and BGD 1071 Phule G 09316 (Collar rot and Vascular wilt) were found dual resistant GLK 2 was found multiple disease resistant

CONCLUSIONS

Studies on **“Succession of soil borne diseases of chickpea and their management through host plant resistance”** led to conclusion that collar rot, vascular wilt and dry root rot incidence was more when the temperature ranged from 25-32⁰ C. However the occurrence of diseases varies from crop growth stages as the collar rot has been observed upto seedling stage only. Wilt symptoms initiated from seedling flowering stage and dry root rot started from Feb. onwards when the temperature rise i e was near to 30⁰ C at pod felling stage. Among 159 entries evaluated 29 entries were exhibited resistant to collar rot, 11 entries resistant to vascular wilt and 19 entries resistant to dry root rot. Entries showing resistance to individual or dual resistance may be utilized in resistance breeding programme after further confirmation.

Suggestions for further work

The entries found resistant to collar rot ,vascular wilt and dry root rot are to be reconfirmed for host resistance to individual as well as dual and multiple resistance .Succession of soil borne diseases of chickpea is to be studied for two consecutive more years to draw the conclusion.

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